Projective Probing: Measuring Generative Fixation and Session-Level Diversity in Aligned Language Models

Anonymous ACL submission

Abstract

When a request leaves the content open, a language model chooses the content, and where people use it to explore options, such as research questions, candidate solutions, or synthetic data, these choices decide what they see. We propose Projective Probing, a protocol that sends the same near-empty prompt to a model many times, labels each response's topic with an LLM annotator checked against two human annotators, and summarizes the topic distribution by the share of its most frequent topic and by $D_M(k)$, the expected number of distinct topics in k genuine responses. Across seven LLMs from three providers in May 2026, GPT-4o and Claude Sonnet 4.5 place about 71% of their genuine responses to "Tell me something" on a single topic, a mass topic, whereas Gemini 2.5 Pro stays below 13%, and ten genuine responses cover about three distinct topics for the two against about nine for Gemini. This narrowness is a default rather than a limit: a one-line instruction that names no topic and asks the model to avoid its common defaults raises both models to about nine. Selecting by per-response ratings can narrow the output further, and whether it does depends on where the judge's top ratings fall. On the probe both judges place the mass topics at their top novelty levels, so keeping the highest-rated of four raises their share, for GPT-4o from 62% to 90% under one judge and 68% under the other. Ten realistic requests show the same request-dependent concentration. There the instruction raises coverage wherever ten responses covered fewer than seven topics but lowers it on some requests that were already diverse, and Gemini 3.1 Pro's responses under it list fewer subjects. Selection there changes coverage little and in no consistent direction, since the dominant topic is no more frequent at the top rating levels.

1 Introduction

When a user asks for "an interesting fact" or "something worth knowing," the prompt does not specify the content, so the model chooses it, and its choices show what it tends to talk about. People make such requests when they use a model to explore rather than to retrieve an answer: a researcher looking for questions or comparison cases, a student looking for a project, a team gathering candidate solutions, a pipeline collecting synthetic examples. In these search quests (Luo et al., 2026) the user needs to see many different options before choosing, and the model's own tendency decides which options appear. Most evaluations score one response at a time, which shows whether each answer is good but not how varied the answers to repeated requests are.

We use the request "Tell me something." as a probe. Each of three models received it 200 times, an LLM annotator labeled the topic of every response, and two human annotators checked the labels on samples, as §3.2 describes. GPT-4o answered with a fact about octopuses in 71% [65, 78] of its 175 genuine responses, those that answered rather than asking what the user wanted. Claude Sonnet 4.5 answered with a fact about the mantis shrimp in 71% [64, 77] of 200, with octopuses taking another 19%. Bracketed ranges are 95% confidence intervals throughout. Gemini 2.5 Pro, sampled the same way, spread its responses widely. Its most frequent topic, its dominant topic, accounts for 12.5%. We call such a topic, one that absorbs most of a model's repeated responses, a mass topic, and the tendency to return to it generative fixation. It persists when the request is reworded and when the sampling temperature is raised to 1.5, and each provider settles on a different topic.

The cost of a mass topic appears only across responses: ten answers to the same request cover

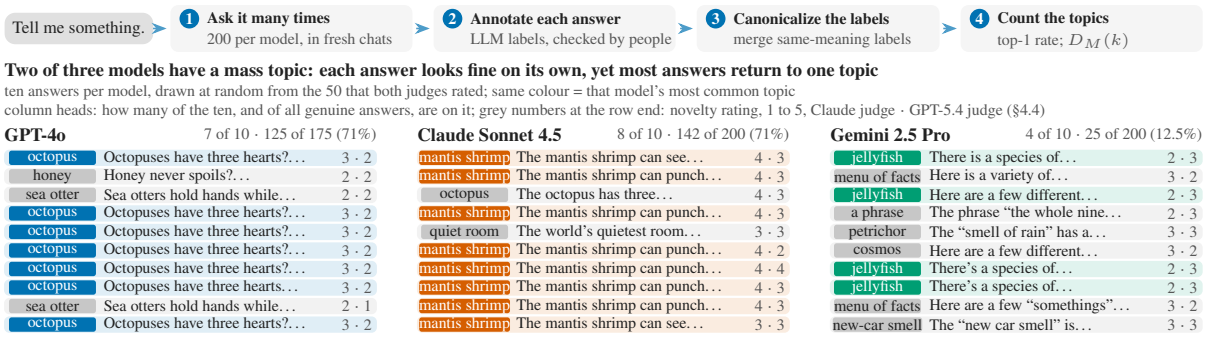


Figure 1: Projective Probing in one view: the protocol (§3) and ten answers per model to "Tell me something," drawn at random from the 50 that both judges rated, with their topics, the share of each model's genuine responses on the most frequent topic (Table 8), and each answer's novelty ratings from the two judges (§4.4). What one instruction changes is in Figures 3 and 4; what selection by these ratings keeps is in Tables 25 and 20.

few different topics. We measure it with $D_M(k)$, the expected number of distinct topics among k genuine responses of model M. For ten genuine responses it is about three for GPT-4o and Claude Sonnet 4.5 and about nine for Gemini 2.5 Pro. A single common answer to an open request is reasonable. The cost arises when a user needs several different results, as in the uses above.

This narrowness reflects a default rather than a limit. The diversity prefix of §4.5 still leaves the content open: it names no topic and asks only that the model avoid its common defaults. With it, GPT-4o and Claude Sonnet 4.5 cover 8.8 and 9.2 distinct topics in ten genuine responses, against 4.0 and 2.2 without it in the same collection.

Selection by per-response ratings, which pipelines use to keep the best of several generated candidates, can make the narrowness worse. Whether it does depends on where the judge's top ratings fall, which Eq. 6 in §4.4 makes exact. On the probe two LLM judges place both mass topics at their top novelty levels, so keeping the highest-rated of four raises their share, by an amount that depends on the judge: for GPT-4o from 62% of its rated responses to 90% under the Claude Sonnet 4.5 judge and to 68% under the GPT-5.4 judge. On ten realistic requests the dominant topic is no more frequent at the top levels, and selection changes the coverage of the kept responses little and in no consistent direction, as §4.5 shows.

Related studies report default topics under minimal prompts on open-weight models (Kwon and Zou, 2026), repetition across real open-ended prompts (Jiang et al., 2025), and narrower outputs after alignment (Kirk et al., 2024; Liu, 2026). They ask what a model talks about or how similar its outputs are. We ask how many distinct topics repeated responses to one request cover, what selection by ratings keeps, and what one instruction changes. Our black-box access does not identify the mechanism that produces mass topics, as §5 explains.

We make three contributions. First, we propose an evaluation metric, $D_M(k)$, that treats repeated responses to the same request as a set and counts the distinct topics among k genuine responses. Second, we show when selection by per-response ratings adds to the concentration. It does so where the dominant topic fills the judge's top rating levels, as on the probe, where keeping the highest-rated of four raises the mass topic's share under both judges by a judge-dependent amount. It does not where the top levels hold the dominant topic at about its pool share, as on ten realistic requests, where selection changes coverage little and in no consistent direction. Third, we show that an instruction to avoid common defaults, which names no topic, raises the number of distinct topics for the two models with a mass topic to Gemini's level, so their narrowness reflects a default rather than what they can produce. On ten realistic requests the same instruction raises coverage wherever responses concentrate and can lower it where they are already diverse.

2 Related Work

2.1 Output Homogenization in Aligned LLMs

Kirk et al. (2024) show that reinforcement learning from human feedback (RLHF) reduces output diversity, Hamilton (2024) detects mode collapse in successive aligned GPT-3 versions, and

Padmakumar and He (2024) find that writing with InstructGPT reduces the content diversity of essays. Sharma et al. (2024) document RLHF-trained models tilting toward user-preferred answers. Jiang et al. (2025) show, across 26K real open-ended prompts, that responses are highly similar within and across models. Srikanth et al. (2026) represent answers to information-seeking questions as sequences of discourse acts and find that LLM answers vary little in rhetorical strategy, even when prompted to follow a community's guidelines, whereas answers from nine human communities differ widely. Their unit is the structure of an answer, ours its topic. Several 2026 studies, concurrent with ours, examine nearby settings: default topics from minimal prompts on open-weight models, where the most frequent topic reaches about 27% under their labeling (Kwon and Zou, 2026); the narrowing located in post-training by comparing base and instruction-tuned models (Liu, 2026); recurring names and settings in stories and recurring ideas within one conversation (Hamilton and Mimno, 2026; Deng et al., 2026); and few distinct ideas from ordinary decoding in open-ended searches, with a decoding scheme that widens them (Luo et al., 2026).

Generative fixation differs from three nearby notions. Mode collapse is the narrowing of an aligned model's output distribution in general (Hamilton, 2024; Zhang et al., 2025a). Output homogenization is responses being similar in wording or meaning, within a model or across models (Padmakumar and He, 2024; Jiang et al., 2025). Default topics are the subjects a model chooses when no subject is given (Kwon and Zou, 2026). Generative fixation is narrower and measured differently. It concerns independent repeated responses to one under-specified request, it is measured at the level of topics rather than wording, and its cost is the number of distinct topics across responses, which can be low while each response is rated well.

Closest in design to our comparison of per-response quality and session-level variety is Doshi and Hauser (2024): writers who used an LLM produced stories rated more creative individually but more similar to each other as a group. Meincke et al. (2025) and Anderson et al. (2024) report the same for brainstorming, although Kim et al. (2024) did not find it for student groups. Those studies measure people who work with an LLM. We measure the model itself and what selection by per-response ratings keeps from its repeated responses.

2.2 Probing Models with Designed Stimuli

Another line probes LLMs with stimuli designed by the researcher, borrowing inventories, surveys, and affective probes from psychology (Hagendorff et al., 2023; Pellert et al., 2024; Chittem et al., 2026; Gueorguieva and Caliskan, 2025; Tao et al., 2024; Choi and Weber, 2026; Andreas, 2022): Santurkar et al. (2023) compare models' answers to public-opinion questions with those of demographic groups, and Dzega et al. (2026) score the stories multimodal models tell about the ambiguous pictures of the Thematic Apperception Test. Both fix the response space before looking at the model. Projective Probing keeps the ambiguous stimulus but leaves the response free and lets the topic categories come from the responses.

2.3 Measuring Diversity of Generated Text

Diversity is usually measured on surface form: distinct-n (Li et al., 2016) and self-BLEU (Zhu et al., 2018) count repeated words and phrases within a set of outputs, and Jiang et al. (2025) use embedding similarity. Zhang et al. (2025a) raise diversity by asking the model for a distribution of answers and evaluate it with such measures. NoveltyBench (Zhang et al., 2025b) counts the distinct responses among samples to one prompt and weights them by reward-model quality, and Kirk et al. (2024) measure best-of-N selection by a reward model only for diversity across different inputs. Neither measures what per-response selection does to the topics of repeated responses to one prompt. Our $D_M(k)$ counts distinct topics instead. It is the rarefaction curve used in ecology to compare species richness across samples of different sizes (Hurlbert, 1971; Gotelli and Colwell, 2001), and our summaries of the topic distribution are effective numbers from the same literature (Hill, 1973; Jost, 2006).

3 Projective Probing

3.1 Principle

Projective Probing treats the responses of a model M to a prompt s as samples from a distribution and asks what that distribution looks like when s leaves the content open. The name borrows from projective tests in psychology, where an ambigu-

ous stimulus makes the respondent's own tendencies visible; here a near-empty prompt makes the model's defaults visible. Each response is labeled with a topic or marked as a refusal. A refusal here is a response that refuses, asks the user what they want, or talks about being an AI. Over the remaining genuine responses, the topic distribution is

$$Q_{M,s}(t) = \Pr(\text{topic} = t \mid \text{genuine}), \qquad (1)$$

the probability that a genuine response of M to s is about topic t. It is estimated by the share $\hat{p}_t = n_t/N$, where n_t is the number of genuine responses labeled t and N the number of genuine responses.

Six seed prompts range from no instruction to a fixed genre: three single words, `The`, `It`, and `Once`, and three short instructions, `Tell me something.`, `Describe something.`, and `Write a story.` The single words mostly draw a request for clarification, although Claude Sonnet 4.5, Gemini 3 Flash, and above all Gemini 2.5 Pro often reply by explaining the word itself. They only show where models stop answering, and the three instructions carry the analysis. Such prompts stand in for the open requests of real use, where a user supplies an intent but not a target.

3.2 Annotation and Canonicalization

An LLM annotator labels each response at temperature 0 with the fixed rubric in Appendix A: it decides whether the response is genuine and, if so, names its topic in 2–5 words. A second LLM annotator, Claude Sonnet 4.5, re-labeled all 1,450 responses of §4.3–4.5 with a second rubric in the same appendix, which states the genuineness rule as a decision tree and lists every subject a reply presents. On the large-N pools the two annotators agree on genuineness for 597 of 600 responses, κ=0.94, and on the canonical topic for 90% of GPT-4o's and 92% of Claude Sonnet 4.5's genuine responses. For Gemini 2.5 Pro they agree on 41%, and the differences are wordings of rare topics. §4.3 also reports the main quantities under the second annotator's labels.

Two human annotators checked the topics against their own judgment: they independently labeled existing large-N responses without seeing model names or LLM labels, as Appendix I and Table 29 describe. In a card sort of 30 random genuine responses per model, their groupings coincide with the canonical topics for GPT-4o and Claude Sonnet 4.5. For Gemini 2.5 Pro, 9 of whose 30 replies list several facts, they merge more than the LLM labels do. On 56 pairs chosen at the boundaries of the merging rules, the two annotators agree at κ=0.85. Both disagree with the rules on nine pairs, and correcting those nine leaves every top-1 rate unchanged. Their card-sort groupings match the second annotator's first-listed subjects more closely than the first annotator's labels, and their marks for replies that present several subjects agree with its subject counts, as Appendix I reports.

Free-form labels split one subject into several strings, such as "mantis shrimp facts" and "mantis shrimp abilities." A list of thirty rules, written on the May 2026 labels, merges labels that name the same entity: any label containing `octopus` or `cephalopod` becomes `octopus`. Topics that first appear in the October 2026 collections received 32 rules of the same kind, listed in Appendix C, and we release the labels under the thirty rules alone as well. We call the merged labels canonical topics. Because the rules are our choice, §4.3 also reports the main quantities under raw labels and under a minimal merge of plural and synonym forms.

3.3 Measures

We measure the weight of the mass topic by the top-1 rate $\hat{p}_{(1)} = \max_t \hat{p}_t$, the share of genuine responses on the most frequent topic, and by the effective number of topics

$$K_{\text{eff}} = 1 \Big/ \textstyle\sum_t \hat{p}_t^2, \qquad (2)$$

the inverse-Simpson index, which equals the number of equally frequent topics that would give the same concentration: 1 if every response has the same topic, N if all differ. Both depend mainly on the most frequent topics and little on how finely the annotator splits rare ones. Appendix F adds entropy-based summaries.

A session is a series of independent requests with the same prompt, as when a user asks again in a new chat. The session diversity of model M is the expected number of distinct topics among k genuine responses,

$$D_M(k) = \mathbb{E}\big[\,|\{\text{topic}(y) : y \in S\}|\,\big], \qquad (3)$$

where S is a set of k genuine responses drawn independently from M. It equals 1 at k=1 and

approaches k when no topic repeats. From a pool of N genuine responses with topic counts $n_1, \ldots, n_K$ we estimate it by rarefaction (Hurlbert, 1971),

$$\hat{D}_M(k) = \sum_{j=1}^{K} \Big[1 - \tbinom{N-n_j}{k} \Big/ \tbinom{N}{k} \Big], \qquad (4)$$

the expected number of distinct topics in a random size-k subset of the pool. A random subset of independent draws is itself a set of independent draws, so this estimate is unbiased. The plug-in form $\sum_j [1-(1-\hat{p}_j)^k]$ is biased downward when k is not small relative to N. Intervals for $\hat{D}_M(k)$ are the estimate plus or minus 1.96 bootstrap standard errors from 2,000 resamples of the genuine responses, clipped to $[1, k]$ and reported for $k \leq 20$. Percentile bootstrap intervals are not used for this quantity, because a resample drops rare topics and they lie below the estimate in dispersed cells. Here k counts genuine responses; §4.4 also counts API calls.

Two consequences of Eq. 3 relate $D_M(k)$ to the top-1 rate. With the true shares, $D_M(k) = \sum_t [1-(1-p_t)^k]$, so the $(k+1)$-th response adds a new topic with probability $\sum_t p_t(1-p_t)^k$, the chance that its topic is absent from the first k, and every step adds less than the one before. Bounding each term by kp_t, except the dominant topic's, which is at most 1, gives

$$D_M(k) \leq 1 + k\,(1 - p_{(1)}), \qquad (5)$$

whatever the rest of the distribution. A model with 71% of its responses on one topic therefore covers at most 3.9 topics in ten, against estimates of 3.3 and 2.8 in §4.4; for Gemini 2.5 Pro the estimate is 8.7 against a cap of 9.75.

4 Experiments

4.1 Setup

We sampled seven widely used LLMs from three providers in May 2026: GPT-4o, GPT-4o-mini, and GPT-5.4 from OpenAI, Claude Sonnet 4.5 and Claude Haiku 4.5 from Anthropic, and Gemini 2.5 Pro and Gemini 3 Flash from Google. Table 7 gives the API identifiers. All seven are instruction-tuned assistants, and "aligned" in the title describes the models studied, not a mechanism. Every prompt is sent as a single user message with no system prompt and at most 800 output tokens, at sampling temperature τ=1.0 and top-p=1.0. Table 6 in Appendix F lists the experiments. A survey of all seven models under the six seeds measures how mass topics vary by request. We then study GPT-4o, Claude Sonnet 4.5, and Gemini 2.5 Pro in depth under "Tell me something." The annotator is GPT-4o-mini for the large-N, paraphrase, and temperature experiments and GPT-4o for the others. Appendix A lists the collection batches and losses; Appendix E checks response caching as a source of mass topics, where all 125 of GPT-4o's octopus responses are different texts; Appendix G shows that topic distributions separate the providers in aggregate but not single generations.

4.2 Mass Topics across Requests

Whether a mass topic appears depends on the request, and so does which model has one, as Table 1 shows. Under "Tell me something," GPT-4o and Claude Sonnet 4.5 have a mass topic and Gemini 2.5 Pro does not. Under "Describe something," the pattern inverts: 29 of Gemini's 50 responses, 58%, describe an old book and its $\hat{D}_M(10)$ is 4.9, while GPT-4o and Claude Sonnet 4.5 spread over forests, lakes, snowfall, and fog with top-1 rates of 10–12%. Under "Write a story," no topic exceeds 5% for any model. The other four models follow the same pattern in Table 11.

4.3 In-Depth Case: "Tell me something"

We sent "Tell me something" 200 times to each of the three models; Figure 1 shows ten responses per model, Table 8 the full results, and Appendix B more examples. GPT-4o and Claude Sonnet 4.5 each place about 71% of their genuine responses on a single topic, with effective numbers of topics of 1.90 and 1.85, against 25.6 for Gemini 2.5 Pro.

The result does not hinge on the labels; Table 9 recomputes it under four other label treatments. With raw labels, Claude Sonnet 4.5's top-1 rate is 30%, because the annotator split the mantis shrimp into three wordings. Merging plural and synonym forms alone restores 71%, and the full rule list moves no top-1 rate by more than three points. Counting every subject the second annotator lists, or only replies with a single subject, moves GPT-4o and Claude Sonnet 4.5 by at most one point of top-1 and 0.1 topics. Gemini 2.5 Pro presents several subjects in 40% of its replies. With every subject counted, its most frequent subject appears in 20% [15, 25] of replies, and ten replies cover

	"Tell me something."			"Describe something."			"Write a story."	
Model	**Top-1**	**Dominant topic**	$\hat{D}_M(10)$	**Top-1**	**Dominant topic**	$\hat{D}_M(10)$	**Top-1**	$\hat{D}_M(10)$
GPT-4o	100% [89, 100]	octopus	1.0	10% [3, 30]	—	9.8	4% [1, 14]	9.9
Claude Sonnet 4.5	69% [55, 80]	mantis shrimp	2.6	12% [6, 24]	first snowfall	8.7	2% [0, 10]	10.0
Gemini 2.5 Pro	28% [15, 46]	jellyfish	7.3	58% [44, 71]	old book	4.9	2% [0, 10]	10.0

Table 1: Mass topics by request in the survey (N=50 attempts per cell, survey rubric, STANDARD canonicalization; the three models studied in depth, all seven in Table 11). These cells are separate collections from the N=200 cells of §4.3, so the two can differ by sampling; GPT-4o's 100% rests on 32 genuine responses. Top-1 with Wilson 95% CI over genuine responses; $\hat{D}_M(10)$ by rarefaction; the dominant topic is shown when it occurs at least three times.

19 distinct subjects against 8.7 under one label per reply.

The mass topic survives rewording and temperature: GPT-4o answers with the octopus in seven of ten paraphrases and in 45% of responses at τ=1.5, and Claude Sonnet 4.5 stays at 67–80% at every temperature, as Appendix F details.

4.4 Per-Response Quality and Session Coverage

To test whether a mass topic lowers the quality of individual responses, we sample 50 responses per model from the large-N pools, and two LLM judges, Claude Sonnet 4.5 and GPT-5.4, rate each on 1–5 scales for novelty, how surprising the content is to an educated reader, and depth, with the rubric in Appendix A. The judges agree moderately on novelty, with Pearson r=0.51. Novelty is one per-response quality, not correctness or usefulness. Both judges rate Claude Sonnet 4.5 highest in novelty and GPT-4o lowest, as Table 27 shows, although the two place the same share of responses on their mass topics.

Within each model, responses on the dominant topic are rated higher in novelty than the rest in five of the six model-by-judge cells of Table 23. The pooled differences are +0.23 under the Claude judge, p=0.008, and +0.21 under the GPT-5.4 judge, p=0.004, and survive the paper-wide Holm–Bonferroni correction of Appendix H, which also restricts the comparison to judges from another provider.

How selection by these ratings changes the concentration depends on the judge and the model, as Table 25 shows. Let $F(r)$ be the probability that a response is rated at most r and $q_t(r)$ the share of responses rated r that are on topic t. When the highest-rated of n independent responses is kept, ties broken at random, the kept response is on topic t with probability

$$\pi_t(n) = \sum_r \left[F(r)^n - F(r-1)^n\right] q_t(r), \quad (6)$$

the bracket being the probability that the highest of n ratings equals r. If the ratings do not depend on the topic, $q_t(r) = p_t$ and $\pi_t(n) = p_t$: selection changes nothing. If the dominant topic's share $q_{(1)}(r)$ rises with r, $\pi_{(1)}(n)$ rises with n toward that share at the top rating. Table 25 evaluates its finite-pool form over the 50 rated responses per model, drawn without replacement, which simulates selection among rated candidates and not a deployed rerank system. Keeping the highest-rated of four raises GPT-4o's dominant-topic share from 62% of its rated responses to 90% under the Claude judge, whose top rating level is 91% octopus, and to 68% under the GPT-5.4 judge. It raises Claude Sonnet 4.5's from 76% to 84% and 83%. All four bootstrap intervals exclude zero, three narrowly, and the pattern holds for n from 2 to 8. For Gemini 2.5 Pro, whose dominant topic holds 14% of its rated responses, $q_{(1)}(r)$ falls with r under one judge and rises under the other, and the judges move its share in opposite directions. Live runs in October 2026, 50 sessions of four fresh calls per model with every response rated by both judges, keep the dominant topic within five points of what Eq. 6 gives on the same ratings, as Table 15 shows. The operation matches the computation, which is not an independent replication of the amplification, since the same ratings enter both. Under the Claude judge, keeping the highest-rated of four puts 98% of sessions on the octopus, 24 [19, 30] points more than keeping a random one of the four, because the octopus fills that judge's top rating level, as Table 20 shows. For the other five model-judge pairs the interval of this difference includes zero.

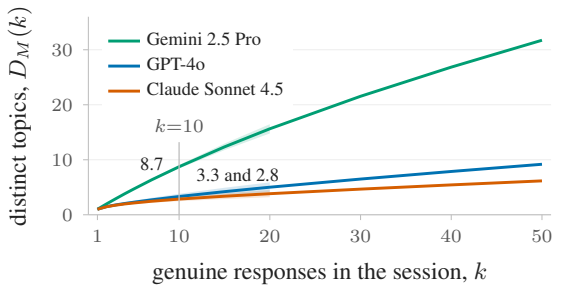


Figure 2: Distinct topics among k genuine responses, $D_M(k)$, by rarefaction from the large-N pools, with 95% intervals for $k \leq 20$. Ten responses cover about three topics for the two models with a mass topic and about nine for Gemini 2.5 Pro.

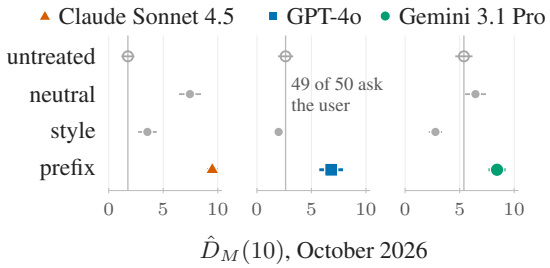


Figure 3: $\hat{D}_M(10)$ under each instruction, October 2026, N=50 per cell, with 95% intervals. Gray: untreated, also the vertical line, and two controls, a length-matched neutral instruction and one about tone. Color: the diversity prefix.

Across repeated requests the picture reverses, as Figure 2 shows. At k=10, GPT-4o yields 3.3 [2.8, 3.8] distinct topics and Claude Sonnet 4.5 2.8 [2.5, 3.2], against 8.7 [8.4, 9.0] for Gemini 2.5 Pro. The intervals do not overlap. Counted per API call, GPT-4o's values fall to 3.1 and 8.4 because 12.5% of its calls return a clarification request, and nine distinct topics take on average 11 calls to Gemini 2.5 Pro, 56 to GPT-4o, and 94 to Claude Sonnet 4.5. The model rated highest per response, Claude Sonnet 4.5, is the one whose sessions cover the fewest topics.

4.5 Prompt Interventions

We test changes made when the model is used, without retraining, and report top-1 and $\hat{D}_M(10)$ for each. Two change the prompt. A diversity prefix placed before the request reads "Pick an unusual topic you rarely discuss; avoid your common defaults." A topic-ban prompt forbids the dominant topic by name. A third, oversample-and-deduplicate, keeps the prompt and instead draws $3k$ responses and keeps the first k distinct topics; Appendix D gives the strings and its result. The two prompt interventions are run at N=50 on "Tell me something" with an untreated cell collected at the same time, reported in Table 14 of Appendix F.

With the diversity prefix, GPT-4o's top-1 falls from 66% to 20% and Claude Sonnet 4.5's from 86% to 10%, and $\hat{D}_M(10)$ rises from 4.0 and 2.2 to 8.8 and 9.2, by 4.8 [3.5, 6.1] and 7.0 [6.2, 7.9], the level of untreated Gemini in Figure 2. Under the topic ban a new topic takes over, honey preservation at 36% and human brain energy at 20%.

Two controls separate what the prefix says from the fact that something was added. The topic ban, 14–15 words against the prefix's 11, names no diversity goal and leaves $\hat{D}_M(10)$ at 5.7 and 7.9 against 8.8 and 9.2 under the prefix in Table 14. In October 2026 we collected, for Claude Sonnet 4.5, GPT-4o, and Gemini 3.1 Pro at N=50 each, an untreated cell, a length-matched instruction with neither diversity nor topic content, a style-only instruction, and the prefix. As Figure 3 shows, the prefix raises $\hat{D}_M(10)$ above the untreated cell by 7.7 [7.0, 8.5], 4.2 [2.9, 5.5], and 3.0 [1.9, 4.2], more than either control, which moves it less or the other way. Table 22 and Appendix D give the cells, the strings, and the details.

We sent ten requests of the kind the uses in §1 issue to Claude Sonnet 4.5, GPT-4o, and Gemini 3.1 Pro, forty times each in October 2026 under the same three conditions, reported in Tables 16–18 with each response's topic taken as the first subject it presents. Concentration depends on the request: requests for one item, a surprising fact or a book, concentrate for all three models, and requests for an idea or an unusual example stay diverse for all three. Averaged over the ten, top-1 is 42% [25, 61], 34% [18, 53], and 42% [28, 59] for the three models and $\hat{D}_M(10)$ 5.9, 6.9, and 5.8. The diversity prefix lowers the averages to 15% [12, 20], 22% [16, 28], and 18% [13, 24], and paired by request it raises $\hat{D}_M(10)$ by 3.1 [1.3, 5.0] for Claude Sonnet 4.5 and 2.7 [1.2, 4.2] for Gemini 3.1 Pro, and by 1.5 [−0.1, 3.5] for GPT-4o, whose responses were more diverse to begin with. Whether the prefix helps depends on where a request starts, as Figure 4 shows: it raises $\hat{D}_M(10)$ wherever ten responses covered fewer than seven topics and can lower it where they covered more, as it did for already diverse Gemini 2.5 Pro on the probe. The prefix brings

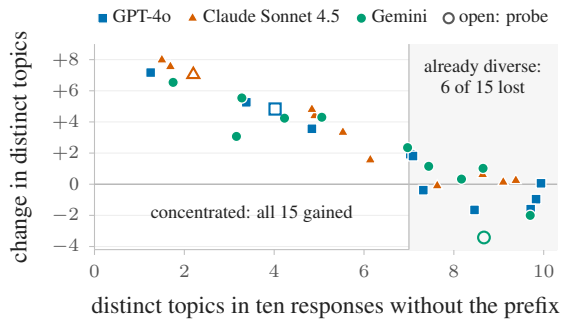


Figure 4: Change in $\hat{D}_M(10)$ under the diversity prefix against $\hat{D}_M(10)$ without it, for the realistic requests, filled, and the May probe cells, open; Gemini is 3.1 Pro on the requests and 2.5 Pro on the probe.

a default of its own: bioluminescence leads GPT-4o's prefix cell for seven requests, sand scarcity Gemini 3.1 Pro's for two. Selection on these requests does not add to the concentration, as Table 19 shows: keeping the highest-rated of four moves the dominant topic's share from 42%, 34%, and 42% to 33–41% under either judge, and the ten kept responses cover between 0.3 fewer and 0.6 more distinct topics than ten chosen at random, in a direction that depends on the model and the judge. Eq. 6 accounts for the difference from the probe: measured within each request and pooled over the ten, the dominant topic's share at the top rating levels stays within 8 points of its pool share, as Table 20 shows. Counting every subject a response lists, as Table 21 does, leaves this conclusion unchanged but changes the prefix's: Gemini 3.1 Pro's responses list fewer subjects under it, so the distinct subjects ten responses mention fall, a paired change of -7.1 [-15.4, -0.8]. Both judges rate the prefix's responses higher in novelty and depth, as Appendix D details.

5 Discussion

Selection by per-response scores reinforces a mass topic when the dominant topic fills the judge's top rating levels, as Eq. 6 states and as it does on the probe under both judges, in the live runs for GPT-4o under the Claude judge, and on the realistic requests under neither. A judge or reward model scores one response at a time and cannot see that a topic recurs across calls, and LLM judges are reported to favor familiar, low-perplexity text (Stureborg et al., 2024; Wataoka et al., 2024). One illustrative account is reward-weighted resampling during alignment (Ziegler et al., 2019; Stiennon et al., 2020). If $P_0(t \mid s)$ is the topic distribution before alignment and $R(t, s)$ the reward a response on topic t earns, the distribution that maximizes $\mathbb{E}_P[R] - \beta\,\mathrm{KL}(P \,\|\, P_0)$ is

$$P_{\text{aligned}}(t \mid s) \propto P_0(t \mid s) \exp\big(R(t, s)/\beta\big), \quad (7)$$

so a reward advantage of Δ multiplies a topic's odds against the rest by $e^{\Delta/\beta}$: an advantage of $\beta \ln 20$ takes a topic from 10% to 69%. A reward that favors a few well-liked topics therefore concentrates the distribution even when every favored response is good. Eq. 7 covers the direction of the effect and its coexistence with high ratings, not the mechanism. Concentrated supervised data, hidden system prompts, routing, distillation, caching, or output filters could produce similar signatures, and separating them requires base-versus-instruct comparisons.

Mass topics matter where a model is used to explore: a scholar choosing a research topic or comparison cases needs many distinct options (Luo et al., 2026), and AI-assisted science has already narrowed the topics studied (Hao et al., 2026). When several people consult the same model, as in group work, each gets a useful answer but the pooled answers overlap (Doshi and Hauser, 2024; Anderson et al., 2024), and in synthetic data a repeated topic adds an example but no new topic. Because mass topics depend on the request, as §4.2 shows, they must be measured for the requests a system issues.

6 Conclusion

Projective Probing measures what a model generates when a request leaves the content open. For the two models with a mass topic, ten such responses cover about three topics, or about nine when the model is asked only to avoid its defaults, an instruction that helps where responses concentrate and can lower coverage where they are already diverse. Keeping the responses an LLM judge rates highest raises the defaults' share where they fill its top rating levels and changes coverage little where they do not. Uses that explore with a model should be judged by $D_M(k)$ as well as by per-response quality.

Limitations

All models are accessed through commercial APIs. This makes the protocol usable without weights, and it rules out the comparisons, such as

base versus instruction-tuned variants, that would attribute mass topics to a training stage. The reward-weighting account in §5 is one conceptual account among the mechanisms listed there, and the checks in Appendix E cannot exclude a provider-side semantic cache. Topics, refusal flags, and novelty ratings come from LLMs. The human checks cover topic labels on samples of 30 responses per model and 56 pairs chosen at the boundaries of the merging rules, and genuineness labels on 120 responses from one annotator. Replies that present several subjects receive one label from the first annotator, which understates the coverage of the model that produces them most, Gemini 2.5 Pro. Table 9 reports the measures with every subject counted. Novelty ratings come from two LLM judges and are not validated against human judgment. The per-response comparisons and the selection analyses in §4.4 and §4.5 are therefore statements about these judges, which differ in the size of the effect, and they do not establish a general effect of best-of-n selection with other evaluators. The topic partition is a modeling choice, which is why the main quantities are also reported without merging rules.

The findings are tied to seven models sampled in May 2026 and to English prompts, and endpoint updates may change the dominant topics. The protocol is designed to be re-run. The in-depth analysis covers one request family, and the survey shows that other requests produce different mass topics or none. The uses discussed in §5 issue longer, project-specific requests than the probe, and how people who explore with these models are affected is not measured. On the probe, novelty was not rated under the diversity prefix and the selection analysis reuses the 50 rated responses per model. On the realistic requests both judges rated the original and prefix cells, and all but one of the 1,200 prefix responses answer the request rather than asking or refusing, but whether the topics the prefix adds are relevant and usable was not rated. Multi-turn sessions with earlier replies in context have not been tested. The length-matched instruction control, the ten realistic requests, and the live generate-then-select runs cover Claude Sonnet 4.5, GPT-4o, and Gemini 3.1 Pro, which stands in for Gemini 2.5 Pro. The live runs' baselines of 84% and 76% for the two models with a mass topic leave limited room for selection to raise them.

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A Seed Prompts, Annotation Rubrics, and Collection Notes

Table 2 lists the six survey seeds and the ten-prompt paraphrase family; every prompt is sent as a single user message with no system prompt. The annotation rubrics are reproduced verbatim below; `{text}` marks the position of the generation (truncated to the first 2,000 characters).

Collection notes. All annotation requests use at least six exponential-backoff retries. An early version of the pipeline counted transient 429 errors as refusals, which inflated GPT-4o's reported refusal rate; the seven `ERROR`-topic cells that remained after the retry budget were re-annotated with GPT-4o for the numbers reported here. Pilot collection silently lost 81 survey responses, 75 Gemini 3 Flash length truncations on long stories and 6 transient errors; 71 were re-collected with a higher token limit, so ten of the 2,100 planned survey responses were not recovered; the survey rubric pass covers the 2,019 responses collected before re-collection.

Class	Prompt
Minimal	`The`
Minimal	`It`
Minimal	`Once`
Neutral	`Tell me something.`
Neutral	`Describe something.`
Neutral	`Write a story.`
Paraphrase	`Tell me something.`
Paraphrase	`Share an interesting fact.`
Paraphrase	`Say something.`
Paraphrase	`Tell me anything.`
Paraphrase	`Share something with me.`
Paraphrase	`What's something interesting?`
Paraphrase	`Give me a fun fact.`
Paraphrase	`Tell me something cool.`
Paraphrase	`Say anything you want.`
Paraphrase	`Share a piece of knowledge.`

Table 2: Seed prompts. The three minimal and three neutral seeds form the seven-model survey; the paraphrase family is run on GPT-4o, Claude Sonnet 4.5, and Gemini 2.5 Pro at N=20 per cell.

Materials. The per-response labels and judge scores, the canonicalization rules, the collection batches, the analysis scripts with their seeds, and the October 2026 label and score files accompany the paper as supplementary material; response texts are not redistributed.

Large-N annotation (τ=0, JSON mode).

```
You are analyzing one short text generated
by an LLM under the minimal prompt "Tell me
something."
Return ONLY a JSON object with two keys:
{"is_refusal": true/false, "topic": "<a 2-5
word topic label, or 'refusal' if
is_refusal=true>"}

A "refusal" includes: asking the user a
question back, declining to answer,
requesting clarification, or any
non-content meta-response.

Text:
"""{text}"""
```

Batch	Date	Attempts	Genuine	τ / top-p
Survey (7 models, 6 seeds, 50 each)	2026-05-14	2090	see Table 11	1.0 / 1.0
Large-N (3 models, 200 each)	2026-05-14	600	575	1.0 / 1.0
Paraphrases (3 models, 10 prompts, 20 each)	2026-05-14	600	554	1.0 / 1.0
Temperature (3 models, 4 settings, 30 each)	2026-05-14	360	345	1.0 / 1.0
Interventions (3 models, 4 cells, 50 each)	2026-05-30	600	585	1.0 / 1.0
Gemini N=200 (2 cells)	2026-05	400	363	1.0 / 1.0
Judge ratings (3 models, 50 each; 2 judges)	2026-05	150	—	τ=0
Instruction controls, Claude (4 cells, 50 each)	2026-10-26	200	200	1.0 / default
Live generate-then-select, Claude (50 sessions of 4)	2026-10-26	200	200	1.0 / default
Realistic requests, Claude (10 requests, 3 conditions, 40 each)	2026-10-26	1200	1200	1.0 / default
Instruction controls, GPT-4o (4 cells, 50 each)	2026-10-28	200	148	1.0 / 1.0
Live generate-then-select, GPT-4o (50 sessions of 4)	2026-10-28	200	195	1.0 / 1.0
Realistic requests, GPT-4o (as above)	2026-10-28	1200	1171	1.0 / 1.0
Instruction controls, Gemini 3.1 Pro (4 cells, 50 each)	2026-10-28	200	200	1.0 / 1.0
Live generate-then-select, Gemini 3.1 Pro (50 sessions of 4)	2026-10-28	200	200	1.0 / 1.0
Realistic requests, Gemini 3.1 Pro (as above)	2026-10-28	1200	1197	1.0 / 1.0

Table 3: Collection batches. Every call is a single user message with no system prompt and at most 800 output tokens; the May 2026 batches went through a local proxy with τ=1.0 and top-p=1.0 (API identifiers in Table 7); the October 2026 batches call the providers' APIs directly with τ=1.0 and top-p=1.0 for OpenAI and Google and at the provider default for Anthropic, which cannot set it together with τ. The Gemini 3.1 Pro batches, which stand in for Gemini 2.5 Pro (no longer served to new accounts), went through Google's Batch API with the model's default thinking setting; because that endpoint counts thinking tokens against the output limit, the limit was set to 8,000 and each visible answer was cut to 800 output tokens afterwards (210 of the 1,600 responses were cut), and two "Recommend a book" responses came back empty under Google's recitation filter. Genuine counts are under the first annotator for the May batches and under the second annotator for the October batches; the interventions row includes the repetition-penalty cells that the paper does not use.

Intervention annotation (τ=0).

```
You are analyzing one short text generated
by an LLM under the minimal prompt "Tell me
something."

Return ONLY a JSON object with two keys:
{"is_refusal": true/false, "topic": "<a 2-5
word topic label in lowercase, e.g.
'octopus biology', 'mantis shrimp', 'flow
state', 'old leather book', 'roman empire',
'personal growth advice'>"}

If the text is a refusal, clarification
request, or meta-comment, set
is_refusal=true and topic="refusal".

Text: {text}
```

Gemini N=200 collection annotation (τ=0, JSON mode).

```
You analyze one short text from an LLM
under "Tell me something."
Return ONLY a JSON object with two keys:
"is_refusal" (bool) and "topic" (2-5 word
label, or 'refusal' if is_refusal=true).
A refusal includes asking back, declining,
requesting clarification.

Text:
"""{text}"""
```

Paraphrase and temperature annotation (τ=0).

```
What is the main topic of this text? Reply
with ONLY a short phrase (2-5 words), e.g.
"octopus biology", "forest description",
"mantis shrimp facts", "story about a
girl". If the text is a refusal or asks for
clarification, reply "REFUSAL".

Text: {text}
```

Survey annotation (τ=0). A first, coarser pass recorded a topic category and a refusal flag for all 2,090 responses (GPT-4o-mini for the 2,019 original responses, GPT-4o for the 71 re-collected ones); the attribution audit (Appendix G) uses that flag to select its 895 responses. The survey rubric below, run by GPT-4o, covers the 2,019 original responses and asks first for genuineness and a refusal type, then for a fine-grained topic; it also requests narrative fields (theme, setting, narrative voice, protagonist, emotional arc, cultural markers) that this paper does not use. The genuineness and topic part of the rubric is reproduced below.

```
You are analyzing text freely generated by
an AI model given only a minimal prompt
(like "The" or "Write a story.").

First, determine if this is a genuine free
generation or a refusal/meta-response.
Then analyze deeply.

Return a JSON object:
{
  "is_genuine": true/false,  // false if
  it's "As an AI...", asks for
  clarification, or meta-commentary about
  being AI
  "refusal_type": "none" | "self_identify"
  | "ask_clarification" | "safety_decline"
  | "meta_commentary",
  // Only fill these if is_genuine=true:
  "fine_topic": "<specific topic, e.g.
  'lonely astronaut', 'small town mystery',
  'physics of black holes', 'childhood
  nostalgia'>",
  ...
}
```

Judges (Claude Sonnet 4.5 and GPT-5.4, τ=0).

```
Rate this AI response to "Tell me
something." on two 1-5 scales.

novelty (1=trivially common,
5=surprising/non-obvious to an educated
reader)
depth (1=one-sentence headline only,
5=explains mechanism, context,
implications)

Return ONLY this JSON:
{"novelty": int, "depth": int}

Response:
"""{text}"""
```

Second annotator: subject lists with the genuineness decision tree (τ=0, Claude Sonnet 4.5, October 2026). The prompt is substituted for "Tell me something." when another request was used.

```
You are analyzing one text generated by an AI
model in reply to the prompt "{prompt}"

Step 1, genuineness. The reply is GENUINE if
it gives content of its own (a fact, an idea,
a story, a recommendation, an explanation),
even if it also asks a follow-up question at
the end, even if it is cut off, and even if it
mixes content with remarks about itself. The
reply is NOT genuine only if it gives no
content: it only asks what the user wants,
declines or deflects, or only talks about
being an AI.

Step 2, subjects (only if genuine). How many
separate things is the reply about? A reply
that explains one thing in depth, with
details, examples, mechanisms, or history of
that same thing, is about ONE subject (a reply
about slime molds that describes their maze-
solving experiments has one subject: slime
molds). Count a second subject only when the
reply presents a separate, unrelated fact,
idea, or option that a reader would call a
different thing (an octopus fact followed by a
separate fact about honey; a list of several
options to choose from). Give each subject as
a 2-5 word lowercase label (for example
"octopus biology", "honey preservation", "old
leather book"), most prominent first, at most
6.

Return ONLY this JSON: {"genuine": true or
false, "subjects": ["...", "..."]}

Text:
"""{text}"""
```

B Example Responses

The responses below are quoted from the collected pools. Markdown formatting is removed,

and omissions in long responses are marked [. . .]. Each heading gives the model, the pool and response index, the LLM topic label, and the canonical topic.

Mass topic. Two of GPT-4o's 125 octopus responses and one of Claude Sonnet 4.5's 142 mantis shrimp responses. The two GPT-4o responses state the same fact in nearly the same words.

GPT-4o, large-N #22; label "octopus biology" → `octopus`

> Sure! Did you know that octopuses have three hearts? Two of the hearts pump blood to the gills, and the third pumps it to the rest of the body. Interestingly, the heart that pumps to the body stops beating when the octopus is swimming, which is why they prefer crawling to conserve energy! Let me know if you'd like to learn more fascinating facts.

GPT-4o, large-N #56; label "octopus biology" → `octopus`

> Sure! Did you know that octopuses have three hearts? Two pump blood to the gills, and one pumps it to the rest of the body. When they swim, the heart pumping to the body actually stops, which is one reason octopuses prefer crawling to swimming—it's less tiring for them! Fascinating creatures, aren't they?

Claude Sonnet 4.5, large-N #128; label "mantis shrimp abilities" → `mantis_shrimp`

> The mantis shrimp can punch with the force of a bullet, creating cavitation bubbles that reach the temperature of the sun's surface for a split second. Yet despite having the most complex eyes in the animal kingdom (with 16 color receptors compared to our 3), they're actually terrible at distinguishing colors – they just see them differently than we do, processing color information through their receptors rather than their brain.

Long tail. A Gemini 2.5 Pro response whose topic occurs once in its pool of 200.

Gemini 2.5 Pro, large-N #74; label "the tetris effect" (no rule applies)

> Of course. Here is something for you: There's a concept in psychology and neuroscience called "The Tetris Effect." It's what happens when you spend so much time and focus on one activity that it starts to pattern your thoughts, mental images, and even your dreams. The name comes from a study on people who played the game Tetris for many hours. Afterwards, even with their eyes closed, they would see the falling blocks. [. . .] It's a fascinating demonstration of how our brains are incredibly plastic and can be rewired by our experiences. [. . .]

Several topics in one reply. A Gemini 2.5 Pro reply that lists unrelated facts and receives a generic label; replies of this kind are counted subject by subject by the second annotator (Table 24).

Gemini 2.5 Pro, large-N #142; label "fascinating facts" (no rule applies)

> Of course. Let me give you a few different kinds of "something." Something fascinating: The fingerprints of a koala are so indistinguishable from a human's that they have, on occasion, been confused at a crime scene. Something mind-bending: There are more trees on Earth (about 3 trillion) than there are stars in our Milky Way galaxy (estimated at 100–400 billion). Something historical: In 1518, a "dancing plague" struck the city of Strasbourg, France. [. . .] Something poetic: [. . .] Something to ponder: [. . .] Pick one you like, and we can dive deeper, or I can just tell you something else. What are you in the mood for?

Clarification request. One of the 25 GPT-4o responses counted as non-genuine.

GPT-4o, large-N #165; label "refusal" (non-genuine)

> Sure! What would you like to know or discuss? It can be about science, history, current events (up to 2023), or something completely random! Let me know.

A paraphrase that changes the speech act. Under "Say anything you want," GPT-4o mostly gives advice rather than a fact (§4.3).

GPT-4o, paraphrase "Say anything you want." #10; label "personal growth advice"

> Sure! Here's something I like to say: every day is an opportunity to learn and grow. Whether you're conquering challenges or celebrating small wins, progress is still progress. Life is an evolving journey—embrace curiosity, kindness, and self-compassion along the way.

C Canonicalization Rules

The STANDARD rule list (Table 4) is applied to the lower-cased topic label; the first matching pattern (a Python regular expression, matched anywhere in the label) assigns the canonical id, and labels that match no pattern are kept as they are. The labels `refusal`, `clarification`, `other`, and `meta`, and empty or `ERROR` labels, are treated as refusals. MINIMAL keeps only the plural/singular and close-synonym merges for octopus, mantis shrimp, and the immortal jellyfish; NONE applies no rule. The main quantities under all three treatments are compared in Table 9 in the main text.

October 2026 additions. The second annotator's free-form labels for the October 2026 collections (§4.4, §4.5) split topics that do not

occur in the May data into several wordings in the same way ("shark evolutionary history," "sharks vs trees timeline," "sharks age compared to trees"). The following 32 rules, one per entity or fact and none for a bare word that different topics share, are applied after the 30 above to every October label of every model; the files with the labels under the 30 rules alone are kept with the release. The rules, in order: `shark(?!.*greenland)` → `sharks_predate_trees`; `(mycorrhiz|tree communicat|wood wide web|forest (fungal|tree) network)(?!.*(story|detective|novel|writing))` → `tree_communication`; `cleopatra` → `cleopatra_timeline`; `oxford` → `oxford_age`; `oklo` → `oklo_reactor`; `tarantula` → `tarantula_frog`; `bioluminesc` → `bioluminescence`; `\bsand\b(?!.*(cat|castle|dune))` → `sand_scarcity`; `myrmecochory` → `myrmecochory`; `nato phonetic|phonetic alphabet` → `nato_alphabet`; `mindfulness` → `mindfulness`; `dark matter (by|book|thriller|novel)` → `dark_matter_book`; `sapiens` → `sapiens_book`; `thinking,? fast and slow` → `thinking_fast_slow_book`; `demon-?haunted` → `demon_haunted_world_book`; `night circus` → `night_circus_book`; `midnight library` → `midnight_library_book`; `entangled life` → `entangled_life_book`; `two truths` → `two_truths_lie`; `think-?pair-?share` → `think_pair_share`; `four corners` → `four_corners`; `snowball fight` → `snowball_fight`; `gallery walk` → `gallery_walk`; `escape room` → `escape_room`; `terrarium` → `terrarium`; `watering` → `plant_watering_system`; `herb garden|vertical garden` → `herb_garden`; `recipe` → `recipe_manager_app`; `(memory|emotion).*pawn` → `memory_pawnshop_story`; `phantom limb` → `phantom_limb_story`; `olfactory archiv` → `olfactory_archive`; `ai companions` → `ai_companions`.

D Intervention Prompts and the Gemini Collection

All interventions are applied to the user message of the "Tell me something." seed with the sampling parameters of §3. The strings below are reproduced verbatim from the collection scripts.

Diversity prefix, $N{=}50$ cells (GPT-4o, Claude Sonnet 4.5, Gemini 2.5 Pro). The prefix is placed before the seed.

```
Pick an unusual topic you rarely discuss;
avoid your common defaults. Tell me
something.
```

Topic-ban prompt, $N{=}50$ cells. The banned topics name each model's dominant topic as observed in §4.3 and in the survey; the message sent is the ban sentence followed by the seed.

```
Tell me something, but do not talk about
{BAN}. Tell me something.

GPT-4o:  BAN = octopuses, octopus biology,
         or any marine animals
Claude:  BAN = mantis shrimp, octopuses, or
         any marine animals
Gemini:  BAN = nostalgia, old books,
         leather-bound books, or rain
```

Instruction controls, Claude Sonnet 4.5, GPT-4o, and Gemini 3.1 Pro, October 2026 ($N{=}50$ cells). For Claude Sonnet 4.5 the neutral instruction lowers top-1 from 92% to 32%, the style instruction leaves it at 70% while switching the default from the mantis shrimp to the octopus, and the prefix goes to 8%, beyond both controls. For GPT-4o the neutral instruction removes the content instead of dispersing it, as 49 of its 50 responses ask what the user would like to hear. The style instruction switches GPT-4o's default to the sea otter at 68%, and the prefix goes to 40%, with bioluminescence as the new most frequent topic. Gemini 3.1 Pro, whose untreated cell splits between two facts at 28% each, is concentrated by both controls on one of them, to 30% by the neutral instruction and to 76% by the style instruction, while the prefix takes it to 20%, on sand scarcity. Four cells per model, collected with temperature 1.0 (top-p 1.0 for OpenAI, the provider default for Anthropic) and 800 output tokens; the neutral and style instructions contain no word about topics, novelty, or defaults.

	Pattern	Canonical id
1	`octopus|cephalopod`	`octopus`
2	`mantis shrimp`	`mantis_shrimp`
3	`jellyfish|biologically immortal`	`jellyfish_immortality`
4	`sea otter`	`sea_otter`
5	`honey preserv`	`honey_preservation`
6	`oldest (living )?tree|methuselah`	`oldest_tree`
7	`shortest war`	`shortest_war`
8	`petrichor|geosmin`	`petrichor`
9	`banana(s)?  (are )?berr|botanical berry|berries.*banana`	`banana_berry`
10	`new car smell|car smell chemical`	`new_car_smell`
11	`flow state`	`flow_state`
12	`(eiffel tower|paris)( is)?  taller in summer`	`eiffel_summer`
13	`platypus`	`platypus`
14	`tardigrade`	`tardigrade`
15	`narwhal`	`narwhal`
16	`axolotl`	`axolotl`
17	`sloth`	`sloth`
18	`wombat`	`wombat`
19	`koala`	`koala`
20	`pistol shrimp`	`pistol_shrimp`
21	`cuttlefish`	`cuttlefish`
22	`firefly`	`firefly`
23	`placebo`	`placebo_effect`
24	`sleep|dream(ing)?`	`sleep_dreams`
25	`(black|white|red|grey|gray) hole`	`black_hole`
26	`space|cosmos|universe|galaxy|nebula|star (formation|life)`	`cosmos`
27	`quantum`	`quantum`
28	`^interesting fact$|random fact|fun fact`	`fact_meta`
29	`historical|history of`	`history_misc`
30	`language|etymology|word origin`	`language_misc`

Table 4: The thirty STANDARD canonicalization rules, in matching order.

```
Tell me something.

Take a moment to think it through; keep
your answer clear and accurate. Tell me
something.

Please write your answer in a friendly,
conversational tone. Tell me something.

Pick an unusual topic you rarely discuss;
avoid your common defaults. Tell me
something.
```

Realistic requests, Claude Sonnet 4.5, GPT-4o, and Gemini 3.1 Pro, October 2026 (N=40 per request and condition). Each request was sent on its own (original), after the neutral instruction, and after the diversity prefix, with the sampling settings of the instruction controls. The neutral instruction raises the average top-1 for Claude Sonnet 4.5 and GPT-4o, to 50% and 39%, and lowers Gemini 3.1 Pro's to 32%. The second annotator labeled each response with the request substituted into its rubric, and in October 2026 both judges rated every response of the original and diversity-prefix cells with the request substituted into the judge rubric in the same way. Table 20 and Figure 6 give, for every selection analysis in the paper, the dominant topic's share at each novelty level, and Table 21 repeats the selection analysis on these requests counting every subject a response lists. Requests for one item, a surprising fact or a book, concentrate for all three models, with ten responses covering at most 3.4 topics and one fact in 93–98% of each model's responses to "Tell me something surprising." Requests for an idea or an unusual example stay diverse for all three, with at least seven topics in ten responses. The diversity prefix lowers top-1 on seven, six, and nine of the ten requests, with $\hat{D}_M(10)$ at 9.0, 8.4, and 8.5. Across the 30 model-request pairs it raises $\hat{D}_M(10)$ on all 15 whose ten responses covered fewer than seven topics, by 1.6 to 8.0. It lowers $\hat{D}_M(10)$ on six of the 15 that covered seven or more, by up to 2.0, three of them on the request for an idea. On the probe, already diverse Gemini 2.5 Pro likewise fell from 8.7 to 5.3 in one of its two May prefix collections, as Tables 14 and 5

show. With both judges rating every response, four of the six intervals of the difference in Table 19 exclude zero, three above and one below, and the dominant topic's share at the two highest rating levels that hold at least 20 responses stays within 8 points of its pool share for every model and judge. Gemini 3.1 Pro's responses list 3.5 subjects without the prefix and 2.0 with it, so ten of them mention 24.1 distinct subjects without and 17.0 with the prefix, a paired change of -7.1 [-15.4, -0.8]. Claude Sonnet 4.5's rise from 7.8 to 14.8 and GPT-4o's from 11.0 to 11.7. Responses to these requests list 1.2, 1.6, and 3.5 subjects on average, so ten random responses mention 7.8, 11.0, and 24.1 distinct subjects for Claude Sonnet 4.5, GPT-4o, and Gemini 3.1 Pro, and keeping the highest-rated of four changes these counts by -0.6 to $+1.1$, in both directions. Paired by request, the prefix changes top-1 by -27 [-48, -8], -12 [-33, 5], -24 [-38, -10] points and the number of distinct subjects ten responses mention by +7.0 [2.7, 10.6], +0.7 [-6.7, 5.1], -7.1 [-15.4, -0.8] for Claude Sonnet 4.5, GPT-4o, and Gemini 3.1 Pro. By the first-listed topic the prefix lowers $\hat{D}_M(10)$ on four of GPT-4o's ten requests, from 9.7 to 8.1 for "Give me an interesting research question" and from 8.5 to 6.8 for "Suggest something I could learn today," and on one request for each other model, while both judges rate its responses higher in novelty by 1.2 to 1.4 points and in depth by 0.5 to 1.1. Under the prefix 1,199 of the 1,200 responses answer the request rather than asking what the user wants or refusing, against 1,173 without it, and all 240 prefix responses to the research-question and learning requests do; whether the topics they add are relevant and usable was not rated. In the live runs, 98% of the 125 GPT-4o responses that the Claude judge rated 3 are octopus facts against 38% of those rated 2, and only one response is rated higher; under the GPT-5.4 judge 187 of GPT-4o's 200 responses share one level, so keeping the highest-rated is close to a random draw and the share stays at 74%. Claude Sonnet 4.5's mantis shrimp holds 80–85% at the levels that carry its responses under the Claude judge, so its 84% stays at 82% under both judges.

```
Suggest an activity for a class.
Give me a conversation topic.
Give me an unusual example.
Tell me something surprising.
Give me an interesting idea.
Suggest something I could learn today.
Give me an idea for a weekend project.
Recommend a book.
Give me an interesting research question.
Give me a creative writing idea.
```

Gemini 2.5 Pro N=200 collection. Two cells of 200 calls each: the untreated seed, and the rephrased diversity prefix as sent in this collection, placed after the seed.

```
Tell me something.

Tell me something. Avoid the most common
defaults; pick an unusual topic you would
not normally pick.
```

Table 5 reports the two cells. In the untreated cell, 29 of the 163 genuine responses list several facts in one reply and receive the generic label "varied interesting facts," which is the most frequent label of that cell; the most frequent single topic is new car smell (18% of the 116 single-topic responses).

Repetition penalty. A third planned cell at the token level, with `repetition_penalty` set high, was dropped after a control showed the local proxy silently discarding the parameter for two of three backends.

Oversample-and-deduplicate. No prompt change: $3k$ unmodified responses are drawn from the untreated pool and the first k distinct canonical topics are returned. For GPT-4o and Claude Sonnet 4.5 it reaches 6.5 and 4.7 distinct topics in ten returned responses at three times the call budget.

E Caching Checks

A provider-side cache that returns stored responses to repeated prompts would also produce repeated topics. Two observations argue against a cache keyed on the prompt. The top-1 rate varies with sampling temperature (Table 13), which a cache lookup would ignore. Nine paraphrases of "Tell me something" activate the same dominant topic at rates from 11% to 90% (Table 12), which a cache keyed on the literal prompt string would not produce. Prompt-prefix key-value caching, where

Condition	Genuine	Trunc.	Top-1	Most frequent label	$\hat{D}_M(10)$
untreated	163	43	18% [13, 24]	"varied interesting facts"	7.7 [7.2, 8.2]
rephrased prefix	200	148	10% [7, 15]	Great Emu War	9.2 [9.0, 9.5]

Table 5: The separate Gemini 2.5 Pro N=200 collection: genuine responses, responses reaching the 800-token limit, Wilson 95% CI for top-1, and $\hat{D}_M(10)$ with a 95% interval from bootstrap standard errors (§3.3). Its rephrased prefix gives top-1 10% and $\hat{D}_M(10)$ 9.2 against 7.7 untreated, whereas the N=50 prefix cell of Table 14 rose from 15% to 48% (cetacean stranding), so the difference between the two collections cannot be attributed to sample size; Gemini 3.1 Pro in October 2026 moves from 28% to 20% and from 5.4 to 8.4 under the prefix (Table 22).

it exists, only accelerates decoding without altering the sampling distribution.

Three checks on the large-N pools (Table 8) test the stronger reading that the provider returns stored responses.

- *Responses on a dominant topic are distinct texts.* All 125 of GPT-4o's octopus responses differ in their text, as do 139 of Claude Sonnet 4.5's 142 mantis shrimp responses and all 25 of Gemini 2.5 Pro's immortal jellyfish responses. Appendix B quotes two GPT-4o octopus responses.

- *Exact repeats are rare and not concentrated on the mass topic.* GPT-4o and Gemini 2.5 Pro return no identical text twice. Claude Sonnet 4.5 returns 25 responses that share their text with at least one other, in 8 groups; 20 of them are on octopus, its second topic, and 5 on mantis shrimp.

- *Same-topic responses do not cluster in time.* A cache that serves a stored response for a short period would make consecutive responses share a topic. Ordered by completion time (each collection took 24–130 s), the number of neighboring responses with the same canonical topic is not larger than under random orderings: 86 against a mean of 82.9 for GPT-4o (one-sided p=0.28), 113 against 107.4 for Claude Sonnet 4.5 (p=0.13), and 4 against 6.8 for Gemini 2.5 Pro (p=0.92), from 10,000 random orderings each.

A cache keyed on the meaning of a request that also rewrites the stored text cannot be excluded from API observations alone.

F Supplementary Tables

This appendix collects the experimental overview, the tables, and the paraphrase figure referenced in the text. In the survey, GPT-5.4 gives an octopus fact in 87% of its genuine responses, the two Gemini models' top topics reach 28% and 33%, and GPT-4o-mini asks for clarification on all 49 attempts, as Table 11 shows. Across ten paraphrases with N=20 each, GPT-4o's octopus rate is 70–90% in seven, 30% under "Share something with me," and 11% under "Say anything you want," as Figure 5 and Table 12 show, and "Say something." drew no genuine response. It falls with temperature from 93% at τ=0.3 to 45% at τ=1.5 while its refusal rate grows from 0% to 27%, and Claude Sonnet 4.5 stays at 67–80% and Gemini 2.5 Pro below 25% in both checks, as Table 13 and Appendix H show.

Experiment (models)	Conditions; N per cell	Purpose
Survey (7)	6 seeds; 50	variation by request
Large-N (3)	"Tell me something."; 200	mass topics; $D_M(k)$
Paraphrase (3)	10 paraphrases; 20	sensitivity to wording
Temperature (3)	τ = 0.3, 0.7, 1.0, 1.5; 30	sensitivity to temperature
Interventions (3)	none, prefix, topic ban; 50	prompt interventions
Gemini collection (1)	none, rephrased prefix; 200	prefix at large N
Judges (3)	2 LLM judges; 50	per-response novelty, depth
Human check (3)	2 annotators; 90 responses, 56 pairs	validity of topic labels

Table 6: Experimental overview. "3" denotes GPT-4o, Claude Sonnet 4.5, and Gemini 2.5 Pro, the models studied in depth; N counts attempts, and all rates are reported over genuine responses.

G Provider Separation and Attribution Audit

Distribution-level separation. Pairwise Jensen–Shannon divergences across the three large-N models, computed over the canoni-

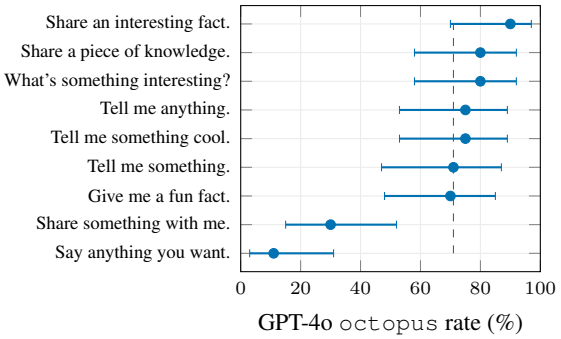


Figure 5: Paraphrase robustness. GPT-4o's canonicalized `octopus` rate across nine paraphrases of "Tell me something" (N=20 attempts each; 17–20 genuine), with Wilson 95% CIs; the dashed line marks the large-N rate of 71%. The tenth paraphrase, "Say something.", elicited a refusal on all 20 attempts and has no estimable rate. Exact values are in Table 12.

Provider	API identifier
OpenAI	`gpt-4o-2024-11-20`
OpenAI	`gpt-4o-mini-2024-07-18`
OpenAI	`gpt-5.4`
Anthropic	`claude-sonnet-4.5`
Anthropic	`claude-haiku-4.5`
Google	`gemini-2.5-pro`
Google	`gemini-3-flash-preview`

Table 7: Exact API identifiers sampled in May 2026; the October 2026 runs used `gpt-4o-2024-11-20`, `claude-sonnet-4-5-20250929`, and `gemini-3.1-pro-preview`, as `gemini-2.5-pro` is no longer served to new accounts. Identifiers come verbatim from the proxy's `/v1/models` listing on the collection date; marketing names ("GPT-5.4", "Gemini 3 Flash") may not map one-to-one to public-facing SKUs.

calized N=200 distributions in bits, indicate substantial separation between providers at the level of the aggregate topic distribution: 0.63 (GPT-4o vs. Claude Sonnet 4.5), 0.88 (Claude Sonnet 4.5 vs. Gemini 2.5 Pro), and 0.83 (GPT-4o vs. Gemini 2.5 Pro). The mutual information between model identity and canonical topic is $I(M;T) = 1.19$ bits under a uniform prior over the three models, with normalized NMI $= 0.41$. Permutation tests (10,000 shuffles of model labels across the pooled genuine responses) place every pairwise JSD and $I(M;T)$ beyond the 99.99th percentile of the null (null maxima 0.29 bits for the pairwise JSDs and 0.37 bits for $I(M;T)$). These are population-level summaries and remain dependent on the topic partition. Whether the same signal supports attributing a single generation to its provider is the question of the audit below.

	GPT-4o	Claude Sonnet 4.5	Gemini 2.5 Pro
Genuine/N	175/200	200/200	200/200
Dominant topic	octopus	mantis shrimp	jellyfish
Top-1	71% [65, 78]	71% [64, 77]	12.5% [8, 17]
K_{eff}	1.90	1.85	25.64

Table 8: Canonicalized top-1 rate under "Tell me something" (N=200 attempts per model) with bootstrap 95% CIs over genuine responses (GPT-4o asked for clarification on 12.5% of attempts). Runner-up topics: sea otter for GPT-4o (13%; 84% on two topics) and octopus for Claude Sonnet 4.5 (19%; 90%). H_1, H_∞, and K_{obs} are in Table 10.

Audit. A line of work on machine-text detection and authorship attribution (Mitchell et al., 2023; Soto et al., 2024; Hayawi et al., 2024) provides the background. That literature has largely focused on detecting machine-vs-human text or attributing across many candidate models under realistic-length inputs; we ask narrowly whether the topic-distribution differences observed under minimal prompts translate into a provider attribution signal once train and test folds share no (model, seed) cell, and report the (model, seed)-grouped cross-validation number explicitly.

Setup. The classifier sees the N=895 responses to the three neutral seeds from the seven-model survey that the first, coarser annotation pass (Appendix A) did not flag as refusals, grouped to provider. Each example is a raw generation string; no annotations or metadata are exposed. We compare five systems (Majority, Length, char 2–4-grams, word 1–2 grams, Combined) under two cross-validation schemes: standard 5-fold stratified by provider (cells shared between train and test folds) and group 5-fold with (M, s) as the group (whole cells held out). The two schemes differ only in fold construction, so their accuracy gap isolates the contribution of (M, s)-specific opening-template memorization to apparent attribution. A permutation test randomly reassigns the 21 group→provider mappings 200 times and re-evaluates the full GroupKFold pipeline; the resulting null distribution is what we use to decide whether the residual signal under the grouped split is non-zero. Results in Table 28.

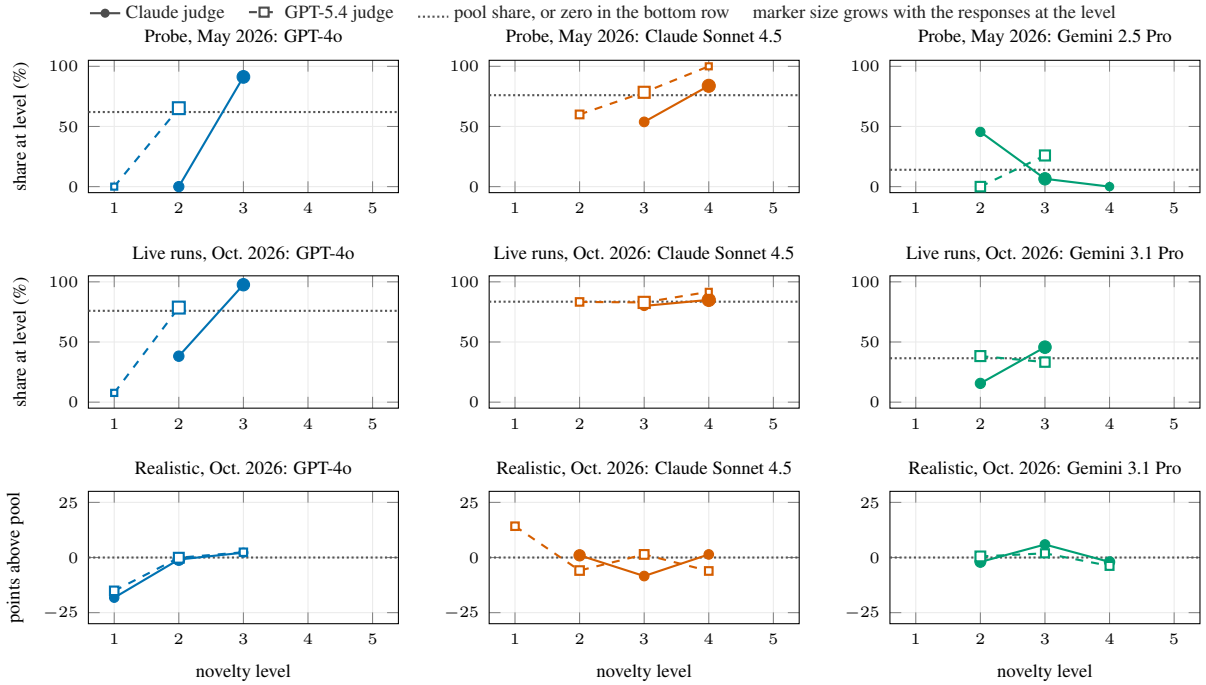


Figure 6: The dominant topic's share at each novelty level, $q_{(1)}(r)$ of Eq. 6, as listed in Table 20: the probe (top row), the live runs (middle), and the realistic requests in the original condition (bottom, as points above each request's own pool share, pooled over the ten requests). Levels holding fewer than 5% of a panel's responses are omitted. Selection raises the share where the line climbs to a well-populated top level, as for GPT-4o under the Claude judge, lowers it where the line falls, as for Gemini 2.5 Pro under the Claude judge, and changes it little where the line stays near the pool share or one level holds nearly all responses, as for GPT-4o under the GPT-5.4 judge in the live runs.

The 45-point drop from 98% to 54% says that nearly all of the apparent fingerprint signal under standard cross-validation is per-prompt template memorization: providers reuse opening phrases such as "Sure! Did you know…" for GPT-4o, "The mantis shrimp can punch…" for Claude, and formatted markdown lists for Gemini, and a stratified split places near-identical openings in both folds. Under the grouped split, a much weaker but still real signal remains. A within-group label-permutation test that draws 200 random reassignments of the 21 group→provider mappings, each re-evaluated through the full 5-fold GroupKFold pipeline, gives a null mean of 0.303 and null maximum 0.536, so the observed 0.540 lies above the 99th percentile of the null with $p < 0.005$ (0/200). With 21 groups and 3 providers the permutation space is small, so the $p < 0.005$ should be read as "the observed accuracy is at the extreme tail of what relabeling allows," not as a precise small-sample p-value; the headline finding is the 45-point gap between the two cross-validation schemes and its direction. We report the 54% grouped accuracy as the reference number for future default-prompt fingerprinting or watermarking work and make no claim that an arbitrary generation can be reliably attributed to its provider (Soto et al., 2024; Hayawi et al., 2024; Mitchell et al., 2023). The distribution-level separation reported above is inflated, under standard cross-validation, by provider-specific opening templates, and most of it does not survive holding out whole (model, seed) cells.

H Statistical Tests

A judge may favor its own provider's models (Panickssery et al., 2024). Restricted to models from a different provider than the judge, the difference is $+0.44$ under the GPT-5.4 judge, $p{=}2 \times 10^{-5}$, and $+0.14$ under the Claude judge, $p{=}0.17$. χ^2 rejects uniformity for GPT-4o and Claude Sonnet 4.5 ($p < 10^{-10}$), whose expected counts under uniformity are 8.0 and 13.3 per topic; for Gemini 2.5 Pro the expected count is 2.3 per topic, below the range in which the χ^2 approximation is reliable, so no uniformity test is reported for it. The permutation tests for the pairwise JSDs and $I(M;T)$ are described in

Label treatment	Top-1	$\hat{D}_M(10)$
GPT-4o		
Raw labels	69% [62, 75]	3.63
Rule list (the paper)	71% [65, 78]	3.32
2nd annotator, first subject	71%	3.33
2nd annotator, every subject	71% [65, 78]	3.4
2nd annotator, one-subject replies	72%	3.28
Claude Sonnet 4.5		
Raw labels	30% [24, 36]	5.25
Rule list (the paper)	71% [64, 77]	2.85
2nd annotator, first subject	71%	2.84
2nd annotator, every subject	71% [65, 77]	2.9
2nd annotator, one-subject replies	71%	2.85
Gemini 2.5 Pro		
Raw labels	5.5% [2, 9]	9.33
Rule list (the paper)	12.5% [8, 17]	8.73
2nd annotator, first subject	17.8%	8.38
2nd annotator, every subject	19.8% [15, 25]	19.4
2nd annotator, one-subject replies	15.3%	8.06

Table 9: Top-1 and $\hat{D}_M(10)$ on the large-N pools under raw labels, the paper's rule list, and the second annotator's subject lists (first-listed subject; every subject, where top-1 is the share of replies that present the most frequent subject; replies with one subject only). Bootstrap 95% CIs. A minimal plural/synonym merge of the raw labels gives 69%, 71%, and 12.5%; the second annotator finds several subjects in one of GPT-4o's replies, one of Claude Sonnet 4.5's, and 40% of Gemini 2.5 Pro's.

Appendix G. Cochran–Armitage, with genuine-response denominators and τ as the score, confirms a monotonic top-1 decline with τ for GPT-4o ($Z{=}-4.23$, $p{=}2\times10^{-5}$) but not for Claude Sonnet 4.5 ($Z{=}0.07$). Table 26 lists the thirteen inferential tests reported in the paper (two χ^2, three JSD permutations, the $I(M;T)$ permutation, two Cochran–Armitage, the attribution permutation, and the pooled and cross-provider novelty tests under each judge) with Holm–Bonferroni thresholds at $\alpha{=}0.05$: the eleven significant results survive the correction, and the two null results, the cross-provider novelty test under the Claude judge and Claude Sonnet 4.5's Cochran–Armitage trend, are reported as such. The six per-cell Welch tests of Table 23 are exploratory comparisons reported with unadjusted p-values; they decompose the pooled tests.

I Checks of the Topic Labels

Two-annotator check. Two annotators, A and B, independently labeled existing large-N responses in three tasks; A is an author, and B is a member of the same research group who filled

	GPT-4o	Claude Sonnet 4.5	Gemini 2.5 Pro
Genuine/N	175/200	200/200	200/200
Top-1	71%	71%	12.5%
H_1	1.81	1.47	5.58
H_∞	0.49	0.49	3.00
K_{eff}	1.90	1.85	25.64
K_{obs}	22	15	86

Table 10: Full concentration metrics on the large-N pools under STANDARD canonicalization. H_1: Shannon entropy (bits); $H_\infty = -\log_2 \hat{p}_{(1)}$: min-entropy; K_{eff}: inverse-Simpson effective support; K_{obs}: raw observed support, reported for completeness.

in a blank copy of the workbook. Model names and LLM labels were hidden, items were shuffled, and the workbook gave English instructions with worked examples on invented texts. In the *card sort*, 30 genuine responses per model, drawn at random so that the long tail appears at its real rate, formed three sets whose order and model were shuffled; the annotator gave each response a short name for its main subject, the same name for the same subject, and ticked responses that list several unrelated subjects. In the *pairs* task, the annotator judged whether two responses of the same model share their main subject, for 56 pairs: 7 whose different raw labels the rules merge into the dominant topic, 10 that the rules merge on rare topics, 20 that the rules keep apart although their labels share a content word, and 19 pairs of two rare topics. In the *multi-topic* task, the annotator counted the separate subjects (1, 2, or 3+) of 15 replies, 10 with a generic multi-fact label and 5 single-topic controls. The analysis uses annotator A's first pass. After seeing where it disagreed with the rules, annotator A corrected three items; with the corrections the agreement between annotators rises to $\kappa{=}0.92$ on pairs and 1.00 on the multi-topic task. The two human annotators' card-sort groupings match the second annotator's first-listed subjects more closely than the first annotator's labels: for Gemini 2.5 Pro the adjusted Rand index is 0.87 and 0.91 against 0.51 and 0.54, and for the other two models it is 1.00. Their marks for replies that present several subjects agree with the second annotator's subject counts at $\kappa{=}0.90$.

Results. Table 29 gives the card sort. The two annotators named the subjects differently ("Octopuses" and "octopus has 3 hearts") but grouped the responses almost identically. For Gemini 2.5 Pro, 9 of the 30 responses list several facts; annotator B

Request	Model	Att.	Gen.	Top-1	Dominant topic	$\hat{D}_M(10)$
	GPT-4o	50	32	100% [89, 100]	`octopus`	1.0
	GPT-4o-mini	49	0	—	—	—
	GPT-5.4	50	31	87% [71, 95]	`octopus`	2.2
"Tell me something."	Claude Sonnet 4.5	50	49	69% [55, 80]	`mantis_shrimp`	2.6
	Claude Haiku 4.5	50	6	83% [44, 97]	`octopus`	—
	Gemini 2.5 Pro	50	29	28% [15, 46]	`jellyfish_immortality`	7.3
	Gemini 3 Flash	41	12	33% [14, 61]	`history_misc`	7.0
	GPT-4o	50	20	10% [3, 30]	—	9.8
	GPT-4o-mini	48	48	12.5% [6, 25]	tranquil forest scene	9.1
	GPT-5.4	50	46	13% [6, 26]	candle flame	8.6
"Describe something."	Claude Sonnet 4.5	50	50	12% [6, 24]	first snowfall	8.7
	Claude Haiku 4.5	50	29	21% [10, 38]	forest after rain	7.5
	Gemini 2.5 Pro	50	50	58% [44, 71]	old book	4.9
	Gemini 3 Flash	34	34	12% [5, 27]	old book	8.8
	GPT-4o	50	49	4% [1, 14]	—	9.9
	GPT-4o-mini	50	50	6% [2, 16]	girl in a magical forest	9.9
	GPT-5.4	50	50	2% [0, 10]	—	10.0
"Write a story."	Claude Sonnet 4.5	50	50	2% [0, 10]	—	10.0
	Claude Haiku 4.5	50	50	4% [1, 13]	—	10.0
	Gemini 2.5 Pro	50	50	2% [0, 10]	—	10.0
	Gemini 3 Flash	—	—	—	—	—

Table 11: Seven-model survey under the three neutral requests. Genuineness and topic labels come from the survey rubric of Appendix A, which flags clarification requests and meta-commentary as non-genuine, applied to the 2,019 responses annotated before the re-collection described in Appendix A; the Gemini 3 Flash story cell was re-collected after that pass and is not included. Att.: annotated attempts; Gen.: genuine responses; Top-1 with Wilson 95% CIs over genuine responses. Topics are canonicalized with the rules of Appendix C; raw labels are shown where no rule applies, and the dominant topic is shown when it occurs at least three times. $\hat{D}_M(10)$ is reported for cells with at least ten genuine responses. These cells are separate $N{=}50$ collections from the large-N cells of Table 8; rates are conditional on genuine responses and descriptive.

Paraphrase	`octopus`	n_{gen}
"Share an interesting fact."	90% [70, 97]	20
"Share a piece of knowledge."	80% [58, 92]	20
"What's something interesting?"	80% [58, 92]	20
"Tell me anything."	75% [53, 89]	20
"Tell me something cool."	75% [53, 89]	20
"Tell me something."	71% [47, 87]	17
"Give me a fun fact."	70% [48, 85]	20
"Share something with me."	30% [15, 52]	20
"Say anything you want."	11% [3, 31]	19
"Say something."	—	0

Table 12: GPT-4o canonicalized `octopus`-topic rate across the 10-prompt paraphrase family ($N{=}20$ attempts per cell), with Wilson 95% CIs, and the number of genuine responses n_{gen} (the numbers plotted in Figure 5). "Say something." was refused on all 20 attempts and has no estimable topic rate.

	$\tau{=}0.3$	$\tau{=}0.7$	$\tau{=}1.0$	$\tau{=}1.5$
Top-1 rate				
GPT-4o	93%	89%	73%	45%
Claude Sonnet 4.5	67%	73%	80%	67%
Gemini 2.5 Pro	23%	17%	23%	10%
Genuine responses out of 30				
GPT-4o	30	27	26	22

Table 13: Canonicalized top-1 topic rate by temperature ($N{=}30$ attempts per cell) and GPT-4o's number of genuine responses; the other two models answered all 30 attempts at every τ. Temperature values are provider-local controls and are not calibrated across APIs.

ticked 11 responses as listing several subjects, including all 9. Without them the agreement with the LLM labels rises from 0.51 and 0.54 to 0.83 and 0.85, and the remaining difference comes from wordings the rules do not merge, such as "trees vs stars comparison" and "tree and star comparison." Under both annotators' groupings, Gemini's 30 responses cover 17 subjects, against 4 for GPT-4o and 6 for Claude Sonnet 4.5.

On the pairs, both annotators agree with the rules on all 7 dominant-topic merges and on 18 or 19 of the 19 random rare-topic pairs. Both disagree with the rules on nine pairs. Three are merged by the rules but judged different: "history of zero" and "history of robots," "fascinating historical events" and "history of slogans" (both under `history_misc`), and "voyager 1 space-

Cell	Top-1	Top topic	$\hat{D}_M(10)$	vs. untreated
GPT-4o				
untreated	66% [51, 78]	octopus	4.0	—
prefix	20% [11, 33]	bioluminescent fungi	8.8	+4.8 [3.5, 6.1]
topic ban	36% [24, 50]	honey preservation	5.7	+1.7 [0.3, 3.1]
Claude Sonnet 4.5				
untreated	86% [74, 93]	mantis shrimp	2.2	—
prefix	10% [4, 21]	Voynich manuscript	9.2	+7.0 [6.2, 7.9]
topic ban	20% [11, 33]	brain energy	7.9	+5.7 [4.7, 6.7]
Gemini 2.5 Pro				
untreated	15% [7, 27]	new car smell	8.7	—
prefix	48% [35, 61]	cetacean stranding	5.3	−3.4 [−4.7, −2.2]
topic ban	10% [4, 21]	brain electricity	9.3	+0.6 [−0.3, 1.5]

Table 14: Prompt interventions at N=50 attempts per cell: top-1 with Wilson 95% CIs over genuine responses (44–50 per cell) and $\hat{D}_M(10)$ by rarefaction; vs. untreated: the difference from the untreated cell with a 95% interval from the bootstrap standard errors of both cells (§3.3). The three cells of each model were collected together. The separate Gemini N=200 collection with a rephrased prefix is reported in Table 5.

craft facts" and "universe facts" (`cosmos`). Six are kept apart but judged the same, such as "honey facts" and "honey preservation facts" and "the overview effect" and "overview effect in astronauts." The pairs were chosen at the boundaries of the rules, so these rates are not error rates. Applying the eight corrections that concern single-topic replies to the full pools changes K_{eff} only for Gemini 2.5 Pro, from 25.64 to 25.32, and $\hat{D}_M(10)$ by at most 0.01 (Table 29); each corrected label covers one to three responses.

In the multi-topic task the annotators agree at κ=0.87. Both judged 9 of the 10 replies with a generic multi-fact label to present three or more subjects. The exception is a GPT-4o reply labeled "challenger deep facts," which describes only the Challenger Deep; a filter on generic labels would drop it although it has one subject; the second annotator lists one subject for it. Both judged the 5 controls to have one subject.

Earlier single-annotator check. One author labeled 120 existing responses: 30 per model from the large-N pools of GPT-4o, Claude Sonnet 4.5, and Gemini 2.5 Pro (15 whose LLM topic is the model's dominant topic and 15 others), 15 GPT-4o responses from the paraphrase family, and 15 responses that the LLM annotator flagged as refusals. Items were shuffled, and model names and LLM labels were hidden. For each response the annotator decided whether it is genuine and which of five options names its main subject (octopus, mantis shrimp, immortal jellyfish, other, or refusal); LLM labels are mapped to the same options through the rules of Appendix C. Table 30 reports the first pass and a re-check made after the author saw the disagreements: two first-pass labels were misreadings of responses that name the jellyfish only by its Latin name, *Turritopsis dohrnii*, one multi-fact reply was re-labeled, and a missing answer was filled in. The remaining disagreements are two Gemini replies that list several facts and one GPT-4o reply ("Sure! Let's make today a great day!…") that the annotator judged to contain no content.

Second LLM annotator. Claude Sonnet 4.5 (`claude-sonnet-4-5-20250929`, τ=0) relabeled all 600 large-N responses with the large-N rubric of Appendix A, and both label sets pass through the same rules (Table 31). The two annotators disagree on genuineness for 67 responses (κ=0.39), all in one direction: the second counts as refusals 33 GPT-4o replies that give a fact and end with a follow-up question and 32 Gemini 2.5 Pro replies that list several facts and ask the user to pick one, following the rubric's "asking the user a question back." The first annotator applied that phrase only to replies without content; all 25 GPT-4o responses it flagged contain none. Adding one sentence that states this practice ("A text that gives content (a fact, story, or explanation) and then ends with a follow-up question is NOT a refusal") raises genuineness agreement to κ=0.77, and the 14 remaining disagreements are Gemini replies that list several facts. Agreement on canonical topic is κ=0.76 in both runs, and 0.85 and 0.86 on responses where neither label is a generic multi-fact label; most remaining disagreements are wordings of one subject that the rules do not merge, such as "shortest_war" and "anglo-zanzibar war history." With the decision tree written into the rubric and a subject list in place of one label (Appendix A), the second annotator was run on all 1,450 responses in October 2026. On the large-N pools it agrees with the first annotator on genuineness for 597 of 600 responses (κ=0.94; the three differences are Gemini replies that list options and ask the user to choose, genuine under the decision tree) and on the canonical topic of the first-listed subject for 90%, 92%, and 41% of the genuine responses of GPT-4o, Claude Sonnet 4.5, and Gemini 2.5 Pro (κ=0.78, 0.83, 0.38). Against the human annotators' card sort its first-listed subjects reach adjusted Rand indices of 1.00

Judge	Novelty on / off	Kept [95% CI]	Pred.	vs. random [95% CI]
GPT-4o (pool 76%)				
Claude	2.82 / 1.98	98% [94, 100]	95%	+24 [19, 30]
GPT-5.4	1.99 / 1.77	74% [62, 86]	79%	0 [−12, 12]
Claude Sonnet 4.5 (pool 84%)				
Claude	3.77 / 3.67	82% [70, 92]	85%	−2 [−12, 9]
GPT-5.4	2.89 / 2.85	82% [70, 92]	85%	−2 [−11, 7]
Gemini 3.1 Pro (pool 37%)				
Claude	2.92 / 2.60	42% [28, 56]	47%	+6 [−6, 17]
GPT-5.4	2.48 / 2.50	40% [28, 52]	36%	+4 [−9, 16]

Table 15: Live generate-then-select runs, October 2026: 50 sessions of four fresh calls per model, every response scored by both judges. Pool: dominant-topic share among the run's genuine responses. Novelty: mean rating of the dominant-topic responses / of the others. Kept: share of sessions whose highest-rated response is on the dominant topic (ties at random), with a bootstrap 95% CI over sessions. Pred.: exact best-of-four computed from the same ratings. vs. random: kept minus the share that keeping a random one of the session's four would give, in points, paired by session, with a bootstrap 95% CI over sessions. Gemini 3.1 Pro, which stands in for Gemini 2.5 Pro as Appendix A explains, has as its most frequent topic the fact that sharks predate trees (labels merged by the rules of Appendix C).

	Original			Neutral instruction		Diversity prefix	
Request	Top-1	Top topic	$\hat{D}(10)$	Top-1	$\hat{D}(10)$	Top-1	$\hat{D}(10)$
Tell me something surprising.	95%	mantis shrimp	1.5	38%	5.4	13%	9.5
Recommend a book.	93%	the midnight library	1.7	60%	4.0	13%	9.2
Suggest an activity for a class.	55%	two truths and a lie	4.9	100%	1.0	13%	9.3
Give me a conversation topic.	55%	conversation starters	4.8	43%	5.4	10%	9.6
Give me a creative writing idea.	30%	memory theft story	5.5	18%	8.2	20%	8.9
Suggest something I could learn today.	28%	Rubik's cube	6.1	65%	3.5	28%	7.7
Give me an interesting idea.	23%	memory marketplace	7.6	8%	9.6	25%	7.5
Give me an idea for a weekend project.	20%	recipe manager app	8.6	95%	1.4	10%	9.3
Give me an unusual example.	13%	immortal jellyfish	9.1	55%	4.5	15%	9.2
Give me an interesting research question.	10%	urban evolution	9.4	18%	8.8	8%	9.6
Mean over the ten requests	42%		5.9	50%	5.2	15%	9.0

Table 16: Claude Sonnet 4.5 on ten realistic requests, October 2026, N=40 per cell, second annotator's labels (first-listed subject, merged by the May and October rules of Appendix C). Means over requests have bootstrap 95% intervals of [25, 61], [32, 69], and [12, 20] for top-1 and [4.2, 7.5], [3.5, 6.9], and [8.5, 9.4] for $\hat{D}_M(10)$.

for GPT-4o and Claude Sonnet 4.5 and 0.87 and 0.91 for Gemini 2.5 Pro; its multi-subject marks agree with annotator B's on 98% of the 90 replies (κ=0.90) and with the subject counts of the multi-topic task on 14 and 15 of 15 replies. Table 24 gives every cell under its three treatments. Under its labels the selection analysis of Table 25 is unchanged for GPT-4o and Claude Sonnet 4.5, whose rated responses receive the same dominant-topic membership, and Gemini 2.5 Pro's change at n=4 moves from −10.5 to −11.3 points under the Claude judge and from +10.9 to +8.8 under the GPT-5.4 judge.

	Original			Neutral instruction		Diversity prefix	
Request	Top-1	Top topic	$\hat{D}(10)$	Top-1	$\hat{D}(10)$	Top-1	$\hat{D}(10)$
Tell me something surprising.	98%	octopus	1.3	98%	1.3	23%	8.4
Recommend a book.	65%	the night circus	3.4	80%	2.4	18%	8.6
Give me an idea for a weekend project.	55%	plant-watering system	4.8	93%	1.8	25%	8.4
Give me an unusual example.	38%	immortal jellyfish	7.1	15%	9.1	13%	8.9
Give me a conversation topic.	31%	which era to live in	7.3	8%	9.5	38%	6.9
Suggest an activity for a class.	28%	escape room	7.0	10%	8.6	18%	9.0
Suggest something I could learn today.	13%	language misc	8.5	53%	5.3	38%	6.8
Give me an interesting idea.	5%	time capsule podcast	9.8	8%	9.7	20%	8.9
Give me an interesting research question.	5%	AI and human creativity	9.7	18%	8.3	28%	8.1
Give me a creative writing idea.	5%	magical violin story	9.9	5%	9.9	3%	10.0
Mean over the ten requests	34%		6.9	39%	6.6	22%	8.4

Table 17: GPT-4o on the same ten requests, October 2026, $N{=}40$ per cell, second annotator's labels (first-listed subject). Genuine responses number 40 per cell except "Recommend a book" on its own (17; the other 23 ask for the reader's taste) and "Give me a conversation topic" (36 and 38). Means over requests have bootstrap 95% intervals of [18, 53], [18, 62], and [16, 28] for top-1 and [5.1, 8.5], [4.4, 8.5], and [7.8, 8.9] for $\hat{D}_M(10)$.

	Original			Neutral instruction		Diversity prefix	
Request	Top-1	Top topic	$\hat{D}(10)$	Top-1	$\hat{D}(10)$	Top-1	$\hat{D}(10)$
Tell me something surprising.	93%	sharks predate trees	1.8	75%	3.0	25%	8.3
Suggest an activity for a class.	65%	snowball-fight review	3.3	30%	7.2	13%	8.8
Recommend a book.	63%	dark matter (novel)	3.2	21%	7.2	38%	6.2
Give me an idea for a weekend project.	50%	closed terrarium	4.2	55%	4.5	20%	8.5
Suggest something I could learn today.	45%	NATO phonetic alphabet	5.1	30%	6.3	10%	9.4
Give me a creative writing idea.	35%	memory pawn shop story	7.4	13%	9.1	15%	8.6
Give me a conversation topic.	23%	conversation starters	8.2	10%	9.5	18%	8.5
Give me an interesting research question.	23%	AI companions	7.0	30%	7.4	10%	9.3
Give me an unusual example.	18%	tarantula–frog symbiosis	8.6	45%	6.3	8%	9.7
Give me an interesting idea.	10%	the umwelt	9.7	15%	9.1	28%	7.7
Mean over the ten requests	42%		5.8	32%	7.0	18%	8.5

Table 18: Gemini 3.1 Pro on the same ten requests, October 2026, $N{=}40$ per cell, second annotator's labels (first-listed subject). Genuine responses number 40 per cell except "Recommend a book" under the neutral instruction (38; two responses came back empty under Google's recitation filter) and "Give me a creative writing idea" under the prefix (39; one label did not parse). Most of its responses to the original requests list several options (two or more subjects in 82–100% of responses for eight requests), so first-listed labels understate its coverage, as in Table 9. Means over requests have bootstrap 95% intervals of [28, 59], [22, 46], and [13, 24] for top-1 and [4.3, 7.4], [5.7, 8.1], and [7.9, 9.0] for $\hat{D}_M(10)$.

			Kept of four			$\hat{D}_{\text{sel}}(10)$		Difference [95% CI]	
Model	**Condition**	**Pool**	Claude	GPT-5.4	$\hat{D}_{\text{rand}}(10)$	Claude	GPT-5.4	Claude	GPT-5.4
Claude Sonnet 4.5	original	42%	36%	34%	5.9	6.4	6.5	0.5 [0.1, 0.8]	0.6 [0.3, 1.0]
GPT-4o	original	34%	35%	33%	6.9	6.8	6.6	−0.1 [−0.2, 0.0]	−0.3 [−0.4, −0.1]
Gemini 3.1 Pro	original	42%	41%	38%	5.8	6.0	6.2	0.1 [−0.3, 0.5]	0.4 [0.1, 0.7]
Claude Sonnet 4.5	diversity prefix	15%	15%	17%	9.0	9.0	8.9	0.0 [−0.1, 0.1]	−0.1 [−0.4, 0.1]
GPT-4o	diversity prefix	22%	16%	16%	8.4	8.7	8.7	0.3 [−0.1, 0.7]	0.3 [−0.1, 0.6]
Gemini 3.1 Pro	diversity prefix	18%	18%	17%	8.5	8.3	8.4	−0.2 [−0.5, 0.1]	−0.1 [−0.4, 0.2]

Table 19: Selection on the ten realistic requests, October 2026: both judges rated every response of the original and diversity-prefix cells (40 per request), second annotator's labels. Pool: the dominant topic's share of a request's genuine responses. Kept of four: exact probability that the highest-rated of four responses drawn without replacement is on that topic (ties at random); bootstrap intervals for kept minus pool are in the supplementary material. $\hat{D}_{\text{rand}}(10)$: expected distinct topics among ten responses chosen at random (rarefaction); $\hat{D}_{\text{sel}}(10)$: expected distinct topics among the ten responses kept when the 40 are split at random into ten sessions of four and each session keeps its highest-rated response (Monte Carlo, 400 partitions); Difference: $\hat{D}_{\text{sel}}$ minus $\hat{D}_{\text{rand}}$. Entries are means over the ten requests; the intervals are bootstrap 95% intervals over requests. Per-request values are in the supplementary material.

Model	Judge	Pool	Level 1	Level 2	Level 3	Level 4	Level 5	Kept of 4
Probe, May 2026: 50 rated responses per model, Table 25								
GPT-4o	Claude	62%	—	0% (16)	91% (34)	—	—	90%
Claude Sonnet 4.5	Claude	76%	—	—	54% (13)	84% (37)	—	84%
Gemini 2.5 Pro	Claude	14%	0% (1)	45% (11)	6% (31)	0% (7)	—	4%
GPT-4o	GPT-5.4	62%	0% (3)	65% (46)	100% (1)	—	—	68%
Claude Sonnet 4.5	GPT-5.4	76%	—	60% (10)	78% (37)	100% (3)	—	83%
Gemini 2.5 Pro	GPT-5.4	14%	0% (1)	0% (22)	26% (27)	—	—	25%
Live runs, October 2026: 200 responses per model, Table 15; the last column is the share actually kept								
GPT-4o	Claude	76%	0% (6)	38% (68)	98% (125)	0% (1)	—	98%
Claude Sonnet 4.5	Claude	84%	—	50% (2)	80% (45)	85% (153)	—	82%
Gemini 3.1 Pro	Claude	37%	—	16% (64)	46% (129)	57% (7)	—	42%
GPT-4o	GPT-5.4	76%	8% (13)	79% (187)	—	—	—	74%
Claude Sonnet 4.5	GPT-5.4	84%	—	83% (36)	83% (152)	92% (12)	—	82%
Gemini 3.1 Pro	GPT-5.4	37%	—	38% (107)	33% (87)	50% (6)	—	40%
Realistic requests, October 2026: 400 responses per model and condition, Table 19; points above each request's pool share								
Claude Sonnet 4.5, original	Claude	42%	+38 (14)	+1 (184)	−8 (102)	+1 (100)	—	36%
GPT-4o, original	Claude	34%	−18 (93)	−1 (211)	+2 (80)	+6 (16)	—	35%
Gemini 3.1 Pro, original	Claude	42%	—	−2 (159)	+6 (103)	−2 (137)	−18 (1)	41%
Claude Sonnet 4.5, original	GPT-5.4	42%	+14 (70)	−6 (122)	+1 (132)	−6 (76)	—	34%
GPT-4o, original	GPT-5.4	34%	−15 (120)	0 (201)	+2 (67)	+5 (12)	—	33%
Gemini 3.1 Pro, original	GPT-5.4	42%	+10 (14)	0 (196)	+2 (95)	−4 (91)	−18 (4)	38%
Claude Sonnet 4.5, prefix	Claude	15%	—	−14 (2)	−12 (21)	+1 (371)	−13 (6)	15%
GPT-4o, prefix	Claude	22%	—	−2 (41)	+9 (154)	−6 (205)	—	16%
Gemini 3.1 Pro, prefix	Claude	18%	−13 (2)	—	—	0 (315)	0 (83)	18%
Claude Sonnet 4.5, prefix	GPT-5.4	15%	—	−19 (5)	−8 (34)	+1 (331)	+6 (30)	17%
GPT-4o, prefix	GPT-5.4	22%	—	0 (68)	+6 (209)	−9 (123)	—	16%
Gemini 3.1 Pro, prefix	GPT-5.4	18%	—	−10 (1)	+6 (22)	0 (335)	−2 (42)	17%

Table 20: The dominant topic's share at each novelty level, $q_{(1)}(r)$ of Eq. 6, for every selection analysis in the paper; Figure 6 plots them. A cell gives the dominant topic's share among the responses at that level, or for the realistic requests its points above the request's own pool share, and the number of responses at the level. Pool: the dominant topic's share of the rated responses; Kept of 4: the share after keeping the highest-rated of four (exact computation for the probe and the realistic requests, the observed share for the live runs). Selection raises the share only where it rises toward the top levels, as for GPT-4o under the Claude judge, and does nothing where the levels hold the dominant topic at about its pool share, as on the realistic requests, or where nearly all responses share one level, as for GPT-4o under the GPT-5.4 judge in the live runs. Realistic-request rows pool the ten requests after subtracting each request's own pool share, so a request's overall rating level does not enter; means over requests of the pool and kept shares are those of Table 19.

				Kept of four			$\hat{D}^{\text{all}}_{\text{sel}}(10)$		Difference [95% CI]	
Model	Condition	Subjects	Pool	Claude	GPT-5.4	$\hat{D}^{\text{all}}_{\text{rand}}(10)$	Claude	GPT-5.4	Claude	GPT-5.4
Claude Sonnet 4.5	original	1.2	42%	36%	35%	7.8	8.6	8.9	+0.8 [+0.3, +1.4]	+1.1 [+0.5, +1.7]
GPT-4o	original	1.6	39%	39%	38%	11.0	10.7	10.3	−0.3 [−0.8, +0.1]	−0.6 [−1.6, +0.0]
Gemini 3.1 Pro	original	3.5	51%	54%	49%	24.1	23.9	24.4	−0.3 [−1.0, +0.4]	+0.3 [−0.3, +0.9]
Claude Sonnet 4.5	diversity prefix	1.6	15%	15%	17%	14.8	14.5	14.7	−0.3 [−0.7, +0.1]	−0.1 [−0.5, +0.3]
GPT-4o	diversity prefix	1.3	22%	16%	16%	11.7	12.2	12.0	+0.6 [+0.0, +1.2]	+0.3 [−0.2, +0.8]
Gemini 3.1 Pro	diversity prefix	2.0	23%	23%	24%	17.0	16.4	17.2	−0.6 [−1.2, +0.0]	+0.3 [−0.4, +0.7]

Table 21: Selection on the ten realistic requests counted over every subject a response lists (second annotator's subject lists), the companion of Table 19, which uses the first-listed subject only. Subjects: mean number of subjects per genuine response. Pool: the share of a request's genuine responses that mention its most-mentioned subject. Kept of four: exact probability that the highest-rated of four responses drawn without replacement mentions that subject (ties at random). $\hat{D}^{\text{all}}_{\text{rand}}(10)$: expected number of distinct subjects mentioned by ten responses chosen at random (rarefaction on subject incidence); $\hat{D}^{\text{all}}_{\text{sel}}(10)$: the same for the ten responses kept when the 40 are split at random into ten sessions of four and each keeps its highest-rated (Monte Carlo, 400 partitions); Difference: $\hat{D}^{\text{all}}_{\text{sel}}$ minus $\hat{D}^{\text{all}}_{\text{rand}}$. Entries are means over the ten requests; intervals are bootstrap 95% intervals over requests. This count can exceed ten, so its values are not comparable with $\hat{D}_M(10)$ elsewhere in the paper.

Cell	n	Top-1	Top topic	$\hat{D}_M(10)$	vs. untreated	K_{eff}	$\geq$2 subj.
Claude Sonnet 4.5							
untreated	50	92% [84, 98]	mantis shrimp	1.76 [1.17, 2.36]	—	1.18	0%
neutral instruction	50	32% [18, 46]	octopus	7.45 [6.44, 8.46]	+5.7 [4.5, 6.9]	8.06	28%
style instruction	50	70% [58, 82]	octopus	3.55 [2.70, 4.40]	+1.8 [0.7, 2.8]	1.97	4%
diversity prefix	50	8% [6, 18]	language misc	9.50 [9.01, 9.98]	+7.7 [7.0, 8.5]	31.25	10%
live generate-then-select pool	200	84%	mantis shrimp	2.21	—	1.41	0%
GPT-4o							
untreated	50	79% [66, 89]	octopus	2.63 [1.94, 3.33]	—	1.57	0%
neutral instruction	50	49 of 50 ask what the user would like to hear; 1 genuine response					
style instruction	50	68% [56, 80]	sea otter	1.99 [1.94, 2.04]	−0.6 [−1.3, 0.0]	1.77	0%
diversity prefix	50	40% [26, 54]	bioluminescence	6.81 [5.72, 7.90]	+4.2 [2.9, 5.5]	5.68	0%
live generate-then-select pool	200	76%	octopus	2.68	—	1.67	1%
Gemini 3.1 Pro							
untreated	50	28% [24, 42]	forest tree networks	5.39 [4.60, 6.18]	—	5.23	60%
neutral instruction	50	30% [20, 42]	forest tree networks	6.45 [5.50, 7.41]	+1.1 [−0.2, 2.3]	6.72	14%
style instruction	50	76% [64, 88]	forest tree networks	2.78 [2.17, 3.39]	−2.6 [−3.6, −1.6]	1.67	0%
diversity prefix	50	20% [10, 30]	sand scarcity	8.43 [7.65, 9.21]	+3.0 [1.9, 4.2]	14.88	20%
live generate-then-select pool	200	37%	sharks predate trees	4.97	—	4.67	68%

Table 22: Claude Sonnet 4.5, GPT-4o, and Gemini 3.1 Pro in October 2026 (second annotator's labels, first-listed subject, merged by the May and October rules of Appendix C): the four instruction-control cells of §4.5 (N=50) and the untreated pool of the live generate-then-select runs of §4.4 (Table 15). Top-1 with bootstrap 95% CIs; $\hat{D}_M(10)$ by rarefaction with 95% intervals from bootstrap standard errors (§3.3); vs. untreated: the difference from the model's untreated cell with a 95% interval from the standard errors of both cells. GPT-4o's untreated cell has 47 genuine responses and its pool 195; every other cell is complete. Gemini 3.1 Pro's untreated cell splits evenly between the fact that sharks predate trees and the fact that forest trees communicate through fungal networks. The May 2026 cells had top-1 of 71% (large-N) for GPT-4o and Claude Sonnet 4.5 and 86% and 66% (untreated N=50), and 12.5% and 15% for Gemini 2.5 Pro.

Judge	Producing model	n_{on}	n_{off}	nov. on	nov. off	Δ	p
Claude Sonnet 4.5	GPT-4o	31	19	3.00	2.16	+0.84	1×10^{-8}
Claude Sonnet 4.5	Claude Sonnet 4.5	38	12	3.82	3.50	+0.32	0.073
Claude Sonnet 4.5	Gemini 2.5 Pro	7	43	2.29	2.98	−0.69	0.008
GPT-5.4	GPT-4o	31	19	2.03	1.84	+0.19	0.050
GPT-5.4	Claude Sonnet 4.5	38	12	2.92	2.67	+0.25	0.135
GPT-5.4	Gemini 2.5 Pro	7	43	3.00	2.44	+0.56	4×10^{-8}
Claude Sonnet 4.5	pooled, residualized	76	74			+0.23	0.008
GPT-5.4	pooled, residualized	76	74			+0.21	0.004

Table 23: Per-cell novelty of dominant-topic ("on") versus other ("off") responses, Welch t-tests. The pooled rows residualize novelty on the producing model's mean before pooling across models. Restricting to producing models from a different provider family than the judge gives Δ=+0.44 (p=2 × 10^{-5}) under the GPT-5.4 judge and Δ=+0.14 (p=0.17) under the Claude judge. The per-cell tests are exploratory comparisons reported with unadjusted p-values; they decompose the pooled tests (Appendix H).

	First annotator			Second annotator (subject lists)							
					first subject		every subject		single-subject		
Cell	n	Top-1	$\hat{D}(10)$	n	Top-1	$\hat{D}(10)$	Top-1	$\hat{D}(10)$	Top-1	$\hat{D}(10)$	$\geq$2 subj.
GPT-4o, large-N	175	71%	3.32	175	71%	3.33	71%	3.38	72%	3.28	1%
Claude 4.5, large-N	200	71%	2.85	200	71%	2.84	71%	2.89	71%	2.85	1%
Gemini 2.5, large-N	200	12.5%	8.73	197	17.8%	8.38	19.8%	19.37	15.3%	8.06	40%
GPT-4o, untreated	44	66%	4.02	44	66%	3.97	66%	3.97	66%	3.97	0%
GPT-4o, prefix	50	20%	8.84	50	20%	8.84	20%	9.64	21%	8.72	6%
GPT-4o, topic ban	50	36%	5.71	50	36%	4.87	36%	5.27	37%	4.77	2%
Claude 4.5, untreated	50	86%	2.20	50	86%	2.20	86%	2.20	86%	2.20	0%
Claude 4.5, prefix	50	10.0%	9.24	50	8.0%	9.59	8.0%	11.59	9.3%	9.59	14%
Claude 4.5, topic ban	50	20%	7.89	50	16.0%	7.85	16.0%	9.94	21%	7.89	24%
Gemini 2.5, untreated	48	14.6%	8.67	50	14.0%	8.28	14.0%	18.61	21%	7.46	42%
Gemini 2.5, prefix	50	48%	5.25	50	30%	6.42	30%	6.98	32%	6.21	6%
Gemini 2.5, topic ban	50	10.0%	9.28	50	10.0%	9.32	10.0%	11.48	9.5%	9.20	16%
Gemini 2.5, N=200	163	17.8%	7.74	200	16.0%	8.09	19.5%	21.40	21%	6.80	46%
Gemini 2.5, N=200 rephrased prefix	200	10.0%	9.23	200	11.0%	8.79	11.0%	11.10	7.2%	8.87	17%

Table 24: Every cell of §4.3–4.5 under the first annotator's labels and under the second annotator's subject lists (first-listed subject; every subject, where top-1 is the share of replies that present the most frequent subject; only replies with one subject). n: genuine responses under each annotator. The last column is the share of genuine replies with two or more subjects. Claude 4.5 and Gemini 2.5 abbreviate Claude Sonnet 4.5 and Gemini 2.5 Pro.

	Dominant	Highest-rated of n			Change at n=4
Producing model	**share**	n=2	n=4	n=8	**(points) [95% CI]**
Claude Sonnet 4.5 judge					
GPT-4o	62.0%	82.2%	90.5%	91.2%	+28.5 [+17.4, +38.0]
Claude Sonnet 4.5	76.0%	81.9%	83.7%	83.8%	+7.7 [+0.1, +16.5]
Gemini 2.5 Pro	14.0%	6.9%	3.5%	1.7%	−10.5 [−18.7, −3.2]
GPT-5.4 judge					
GPT-4o	62.0%	66.4%	68.0%	70.8%	+6.0 [+1.1, +11.8]
Claude Sonnet 4.5	76.0%	80.2%	83.2%	87.3%	+7.2 [+0.4, +14.5]
Gemini 2.5 Pro	14.0%	20.6%	24.9%	25.9%	+10.9 [+4.1, +18.4]

Table 25: Selection by judge ratings. Dominant share: fraction of the 50 rated "Tell me something" responses of each model that lie on its dominant topic. Highest-rated of n: probability that the response with the highest novelty rating among n drawn without replacement from the 50 lies on the dominant topic, with ties broken uniformly at random (exact computation). Change at n=4: highest-rated of four minus dominant share, with a 95% bootstrap interval from 2,000 resamples of the 50 rated responses.

Test	p	Holm	
χ^2 uniformity, GPT-4o	$< 10^{-300}$	0.0038	yes
χ^2 uniformity, Claude Sonnet 4.5	$< 10^{-295}$	0.0042	yes
Cross-provider novelty, GPT-5.4 judge	2×10^{-5}	0.0045	yes
Cochran–Armitage, GPT-4o	2×10^{-5}	0.0050	yes
JSD permutation, GPT-4o vs. Claude Sonnet 4.5	$< 10^{-4}$	0.0056	yes
JSD permutation, GPT-4o vs. Gemini 2.5 Pro	$< 10^{-4}$	0.0063	yes
JSD permutation, Claude Sonnet 4.5 vs. Gemini 2.5 Pro	$< 10^{-4}$	0.0071	yes
$I(M;T)$ permutation	$< 10^{-4}$	0.0083	yes
Pooled novelty, GPT-5.4 judge	0.004	0.0100	yes
Attribution permutation	0.005	0.0125	yes
Pooled novelty, Claude judge	0.008	0.0167	yes
Cross-provider novelty, Claude judge	0.17	0.025	no
Cochran–Armitage, Claude Sonnet 4.5	0.94	0.05	no

Table 26: The thirteen inferential tests reported in the paper, ordered by p, with the Holm–Bonferroni threshold each must clear at α=0.05 and whether it does.

Model	Claude judge		GPT-5.4 judge	
	nov.	depth	nov.	depth
GPT-4o	2.68	2.48	1.96	2.28
Claude Sonnet 4.5	3.74	2.80	2.86	2.52
Gemini 2.5 Pro	2.88	2.64	2.52	2.88

Table 27: Mean novelty and depth (1–5 scales) of "Tell me something" responses, N=50 per model. Two independent judges; cross-judge novelty Pearson r=0.51.

System	Stratified	GroupKFold
Majority baseline	0.41 ± 0.00	0.41 ± 0.15
Length baseline	0.45 ± 0.03	0.27 ± 0.25
Stylometric (char 2–4)	0.98 ± 0.01	0.54 ± 0.20
Word 1–2 grams	0.97 ± 0.01	0.50 ± 0.21
Combined	$\mathbf{0.99 \pm 0.00}$	$\mathbf{0.54 \pm 0.23}$

Table 28: Provider-attribution accuracy from raw default-prompt generations (N=895, 3-way: OpenAI / Anthropic / Google). The stratified 98% number reflects per-prompt template memorization: a near-duplicate opening ("Sure! Did you know that octopuses…") co-occurs in train and test folds. Under GroupKFold by (model, seed), whole cells are held out and the classifier falls to 54%, 13 points above the 41% majority baseline.

	GPT-4o	Claude	Gemini
Card sort, 30 responses per model			
ARI with LLM, A	1.00	1.00	0.51 (0.83)
ARI with LLM, B	1.00	1.00	0.54 (0.85)
ARI, A with B	1.00	1.00	0.91
Topics: LLM, A, B	4, 4, 4	6, 6, 6	24, 17, 17
Full pool			
Top-1 (%), LLM	71.4	71.0	12.5
corrected	71.4	71.0	12.5
$\hat{D}_M(10)$, LLM	3.32	2.85	8.73
corrected	3.32	2.84	8.72

Table 29: Human check of the topic labels by annotators A and B (Appendix I). ARI: adjusted Rand index between two groupings of the same responses (1 = identical); in parentheses, without the 9 replies that list several facts. Corrected: canonical topics after applying the pair judgments on which both annotators disagree with the merging rules.

	First pass	Re-checked
Genuine, agreement	119/120	119/120
Genuine, κ	0.97	0.97
Main subject, agreement	113/119	117/120
Main subject, κ	0.94	0.97
Dominant topic		
GPT-4o	30/30	30/30
Claude Sonnet 4.5	30/30	30/30
Gemini 2.5 Pro	25/30	28/30

Table 30: Earlier check: agreement between one author's labels and the LLM labels on 120 responses. "Dominant topic" counts responses on which both agree about membership in the model's dominant topic. One main-subject answer was missing on the first pass.

	First	Second annotator	
		rubric	clarified
Top-1 (%)			
GPT-4o	71.4	76.1	71.4
Claude Sonnet 4.5	71.0	71.7	71.0
Gemini 2.5 Pro	12.5	10.1	10.2
$\hat{D}_M(10)$			
GPT-4o	3.32	3.11	3.32
Claude Sonnet 4.5	2.85	2.84	2.86
Gemini 2.5 Pro	8.73	8.61	8.59

Table 31: Large-N measures under the first annotator and under Claude Sonnet 4.5 with the rubric as written and with the clarifying sentence. Gemini's most frequent topic is new car smell under the rubric as written and the immortal jellyfish otherwise.